Trait-based Ecology II

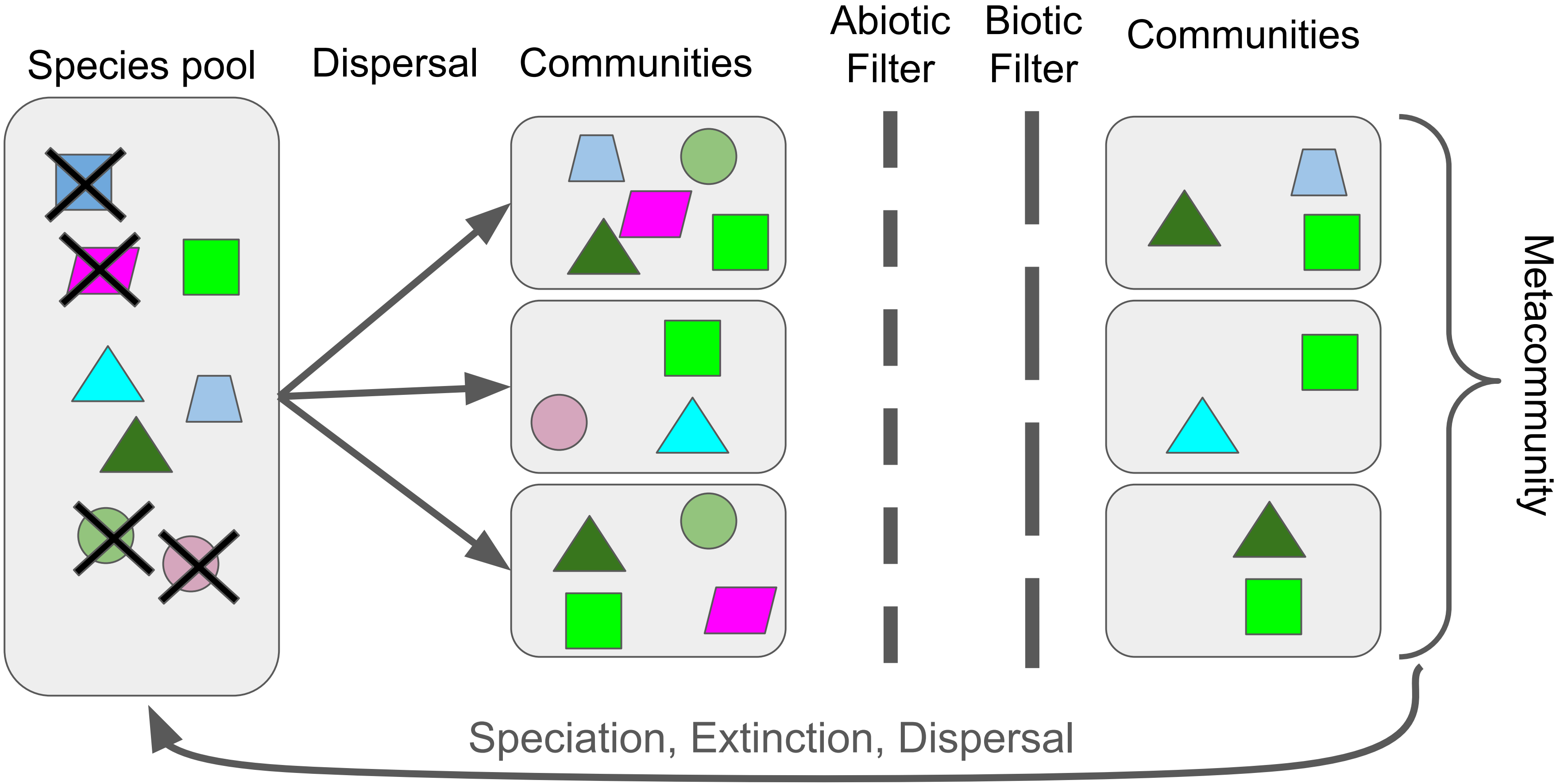
Today's Agenda:

- Quiz
- Project 3: Trait-based ecology
- Trait-based community assembly

Why focus on traits?

- Generality:
    - Rather than focusing on particular species in particular systems, we can say things about all species in all systems (or at least a larger subset of species/systems).
- Quantitative:
    - We move from species (categorical) to continuous measurements
- Links to mechanism
    - We can focus on traits that are linked to particular mechanisms
    - E.g., if we want to work on predation, mouth size might be useful.
- More predictive
    - Allow us to say something about potential future scenarios

Community assembly revisited



Traits and community assembly

Abiotic filtering

Biotic filtering

Dispersal

Traits and community assembly

Abiotic filtering

Biotic filtering

Dispersal

Traits and the abiotic environment

- Environmental filtering
- Trait-environment relationships
- Physiological models of trait-environment relationships

Traits and the abiotic environment

A lot of focus on "environmental filtering" in community assembly

- Species A can/cannot survive in a given climate
- Allows us to say something about species A

Traits allow us to go further!

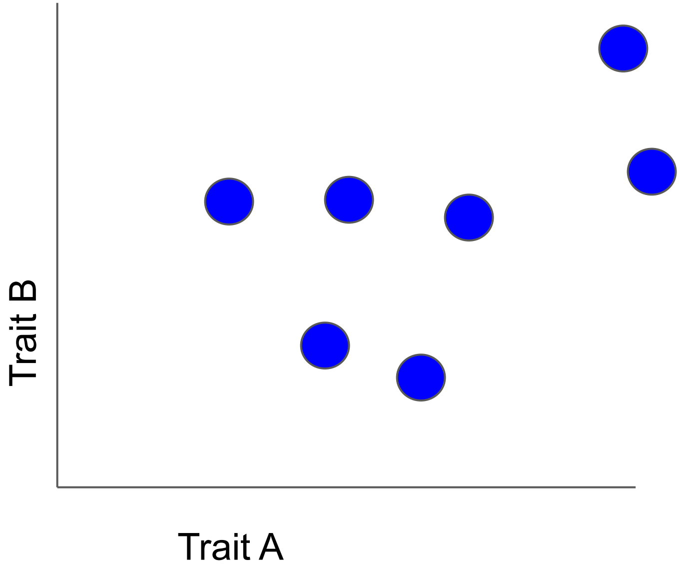
- These are the traits that allow a species to survive in an environment
- Allows us to make predictions across ALL species (with trait data)
- Help us understand WHY species do well/poorly

Traits and the abiotic environment

Different approaches

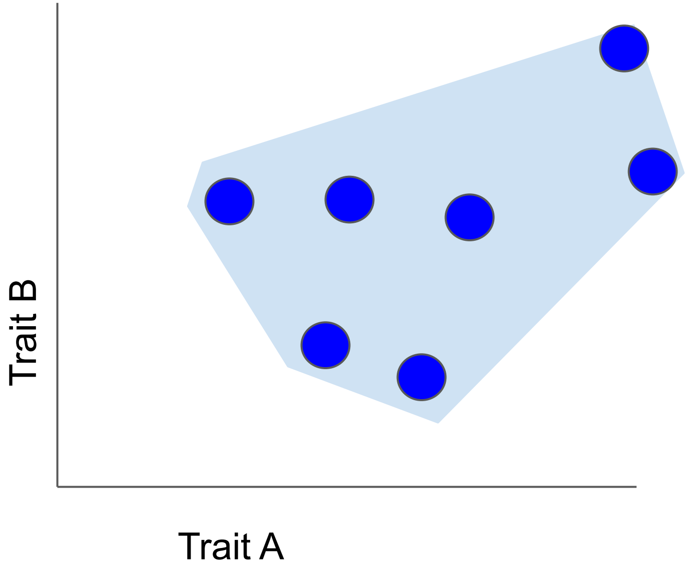
- Pattern-based: e.g., environmental filtering should decrease variance
- Correlation-based: e.g., body size increases with temperature
- Mechanistic: e.g., fancy models that explicitly link traits to performance

Functional Metrics



Functional Metrics

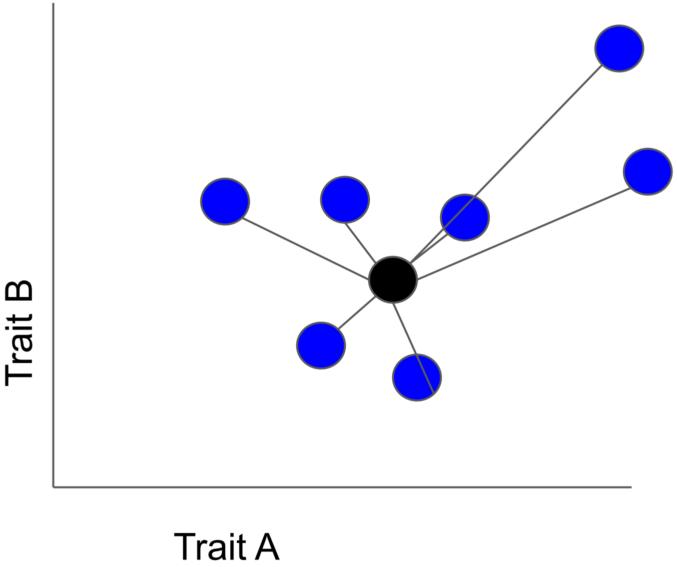
Functional Richness: Total area occupied in trait space



Functional Metrics

Functional Richness: Total area occupied in trait space

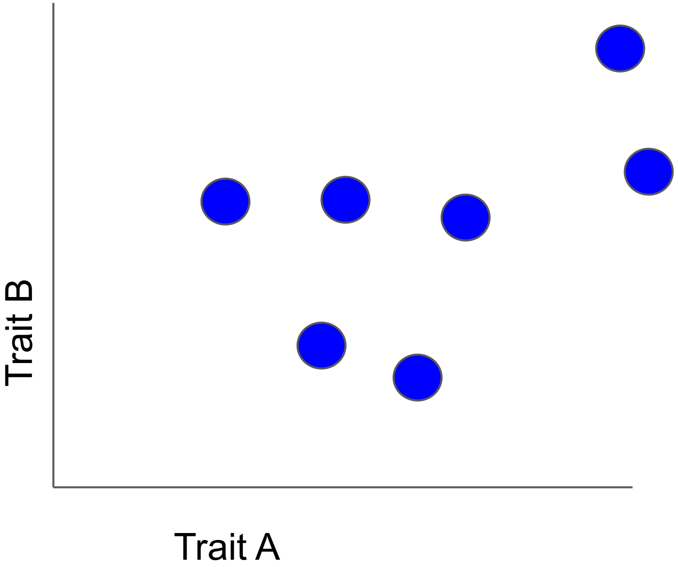
Functional Dispersion: (weighted) Mean distance between species and centroid



Functional Metrics

Richness: How much diversity?

Dispersion: How different are species?

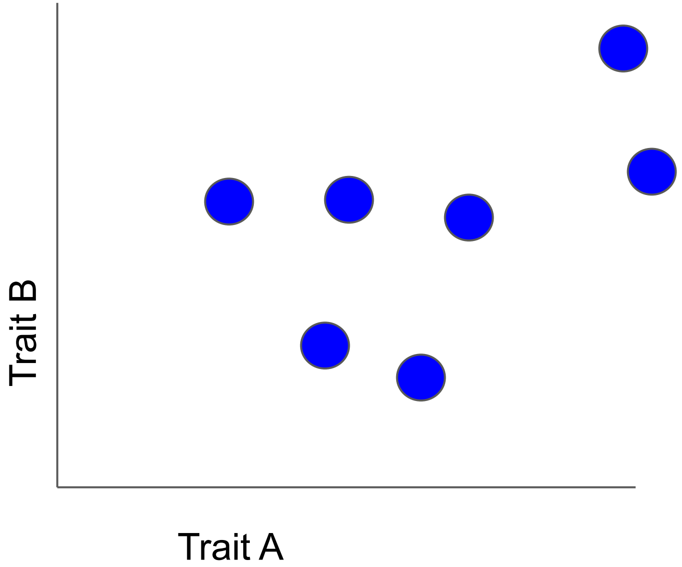


Functional Metrics

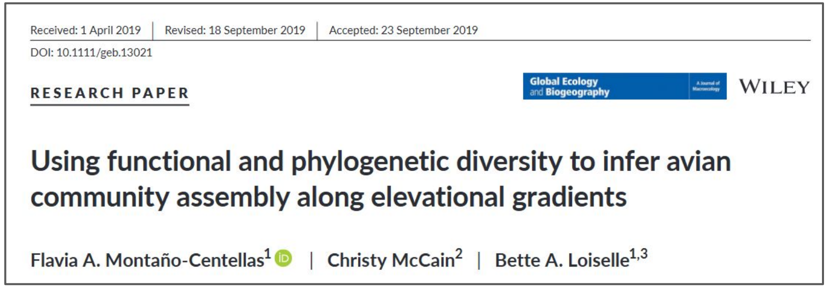
Richness: How much diversity?

Dispersion: How different are species?

How will environmental filtering impact these metrics?



Traits and filtering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RESEARCH PAPER

Global Ecology and Biogeography
WILEY

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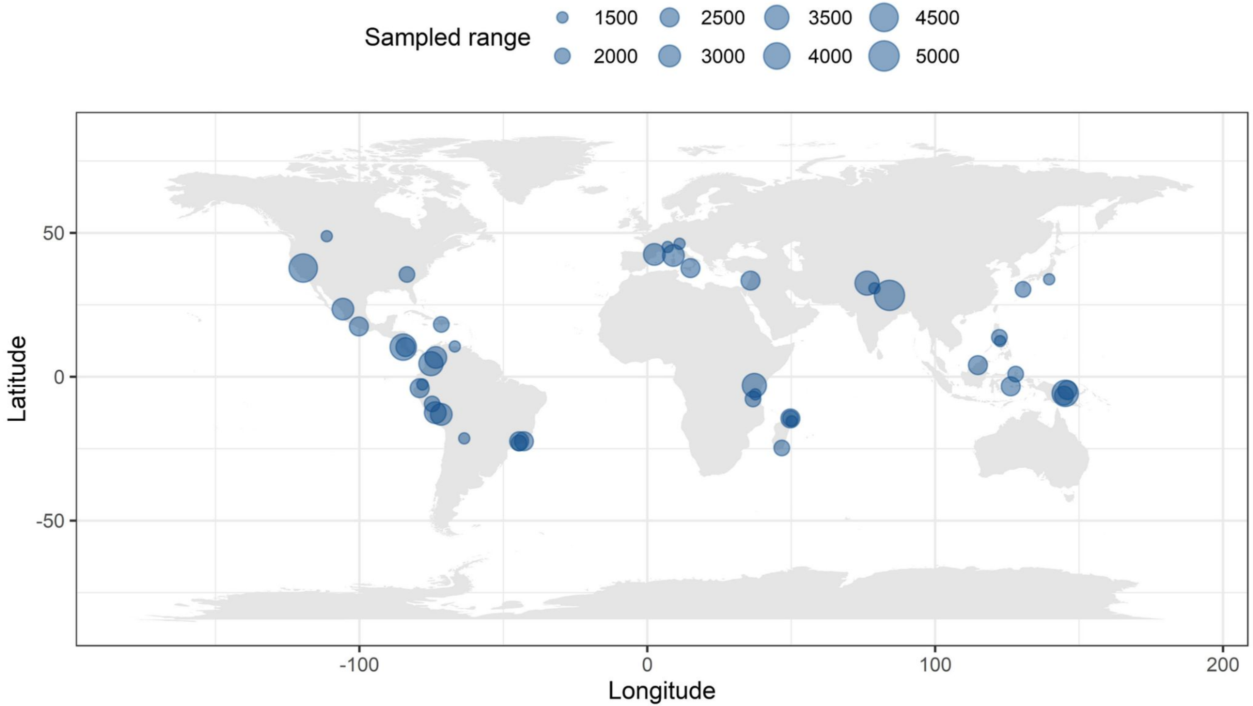
Flavia A. Montaño-Centellas[1] | Christy McCain[2] | Bette A. Loiselle[1,3]



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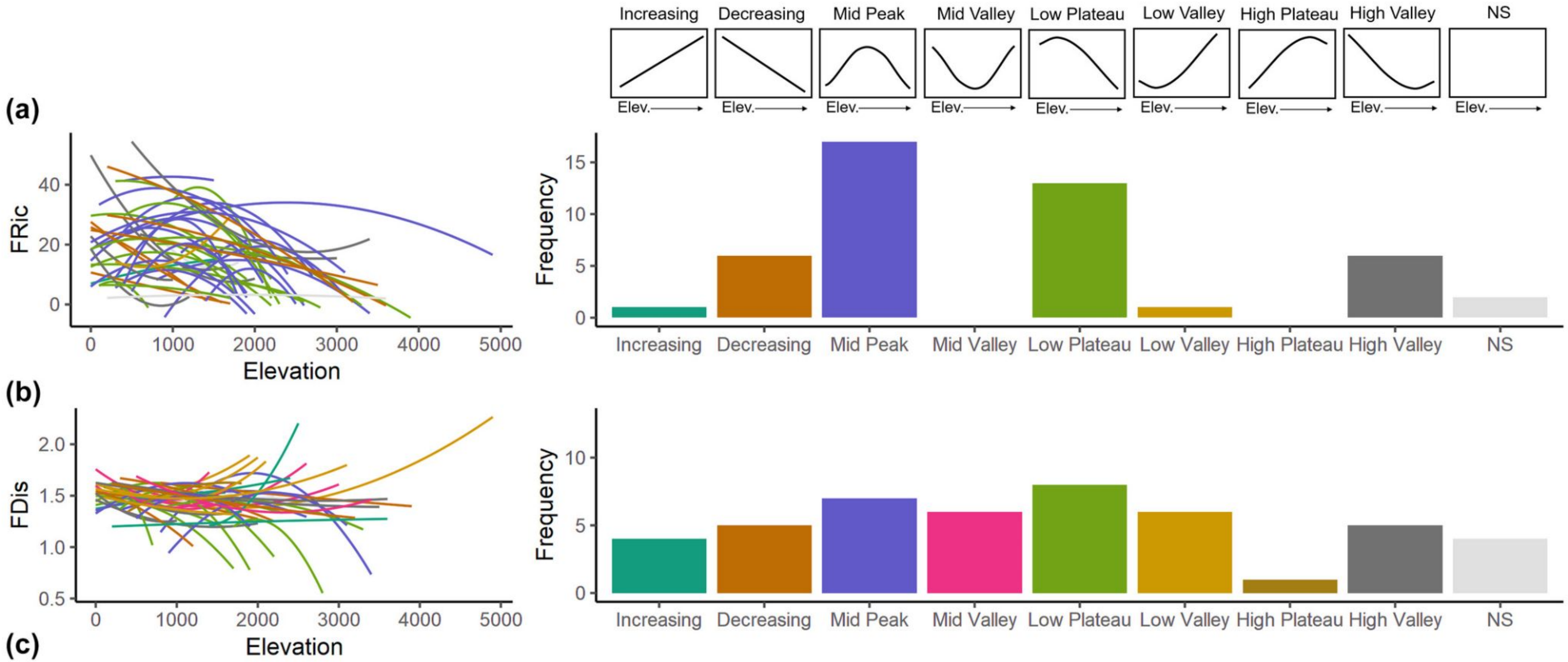

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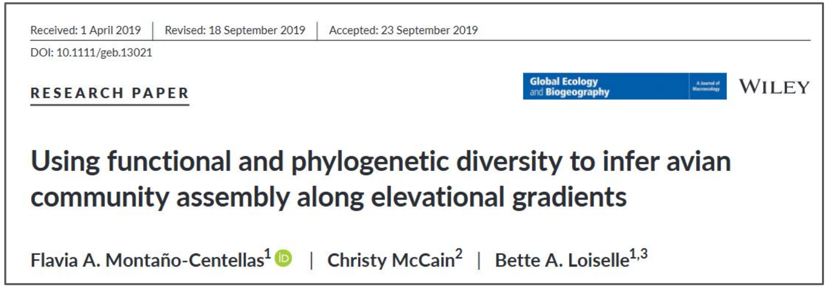
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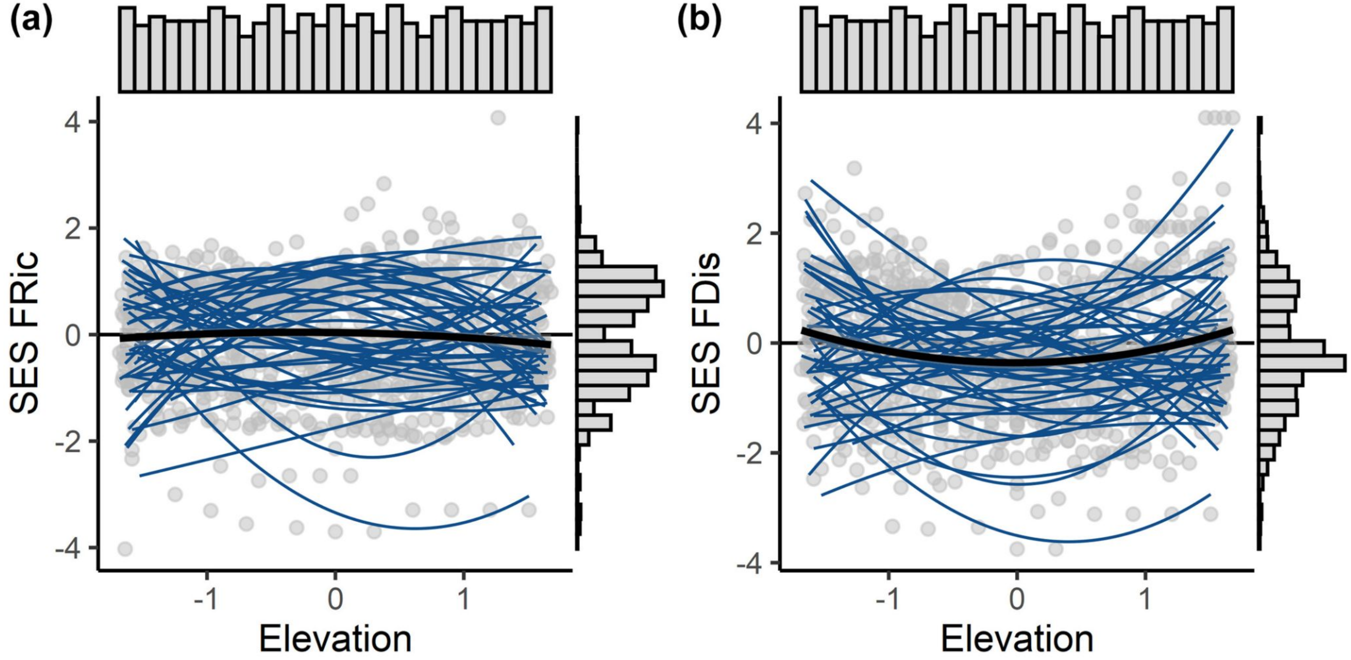

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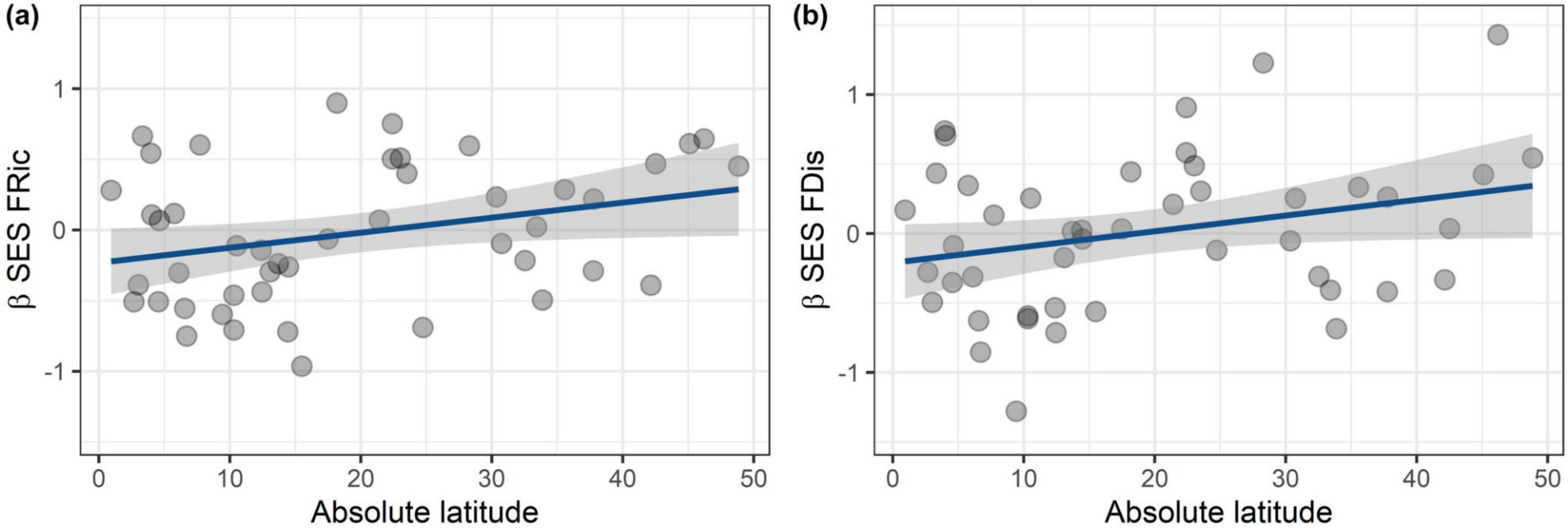

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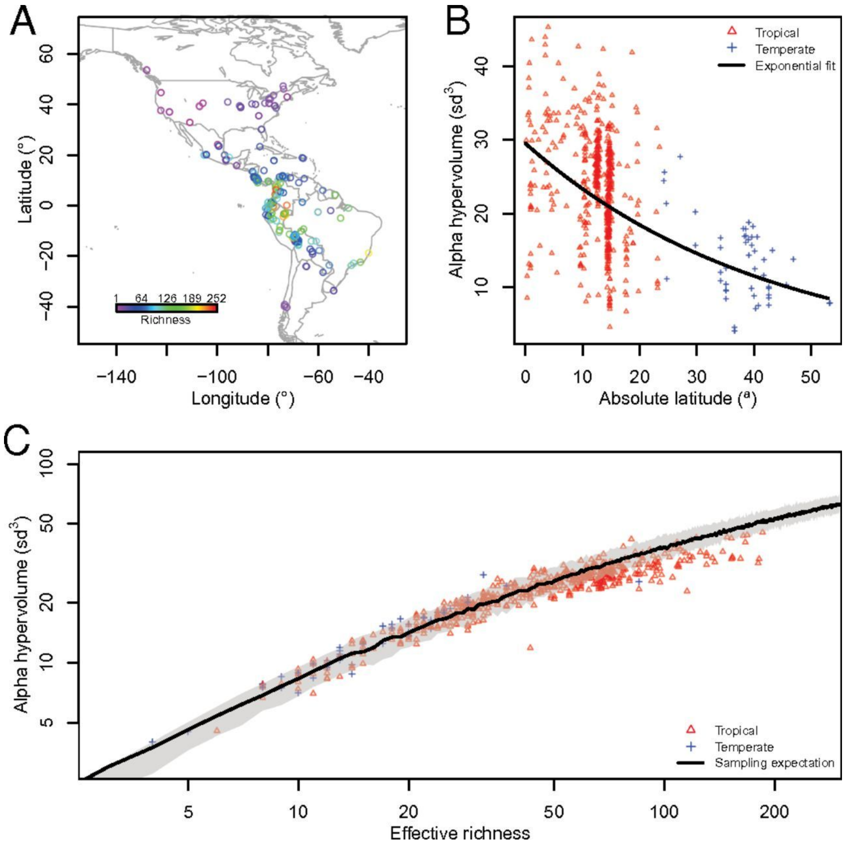

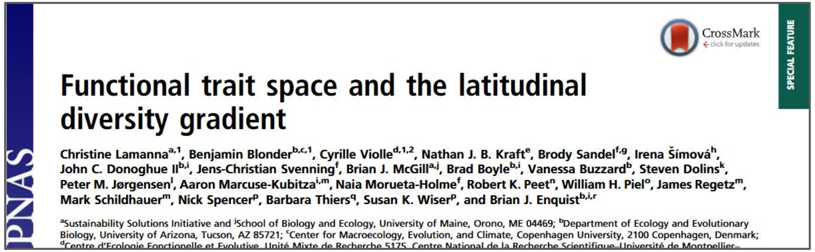
PNAS

SPECIAL FEATURE

CrossMark

Functional trait space and the latitudinal diversity gradient

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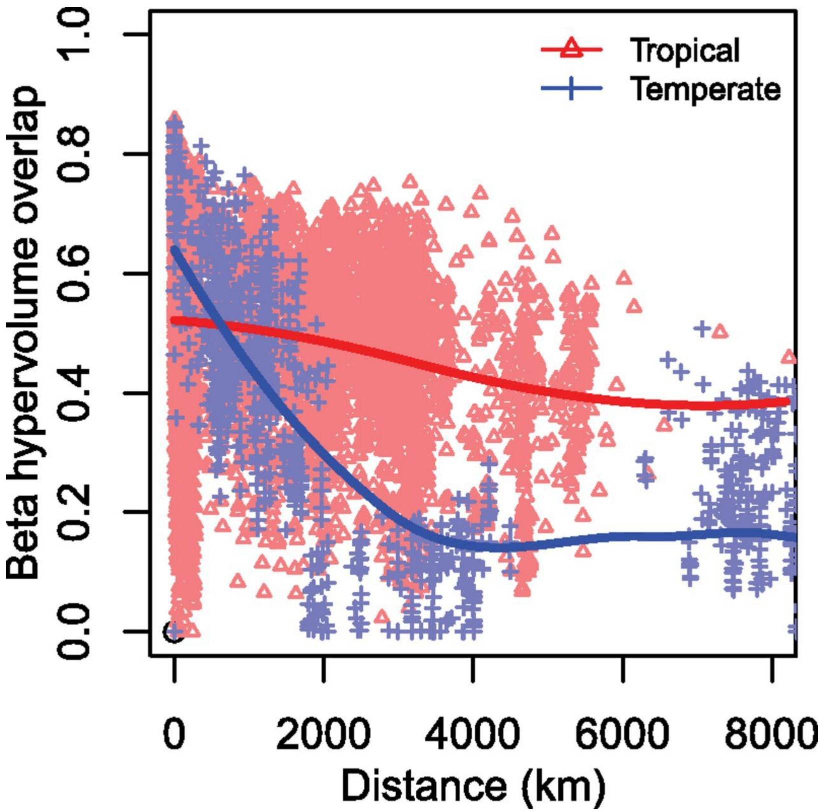

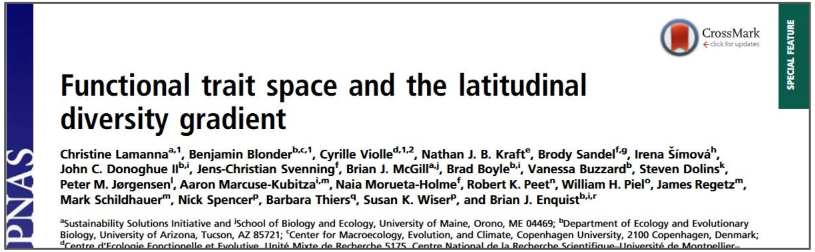

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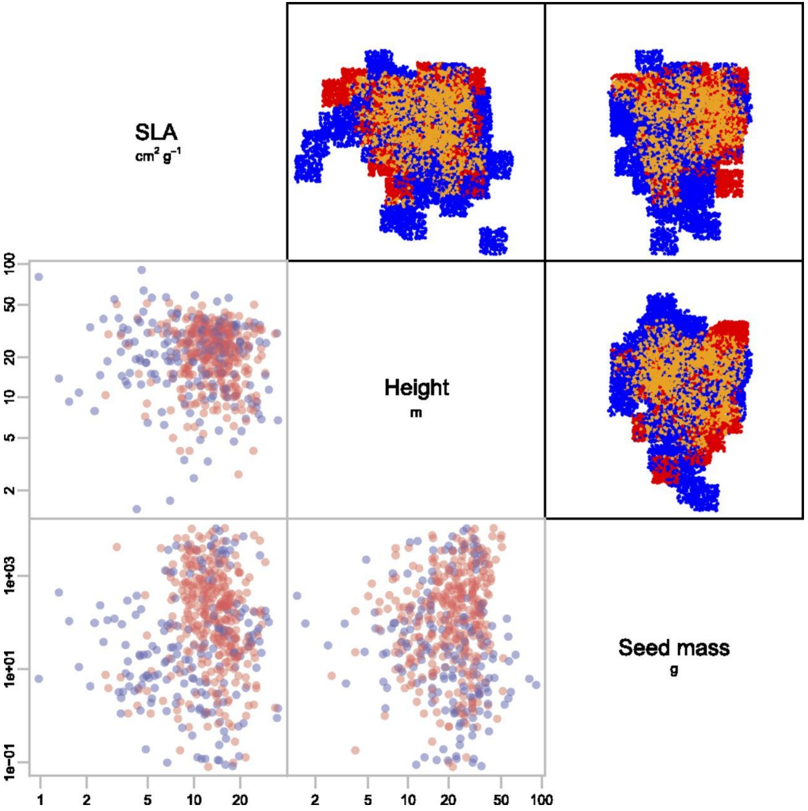

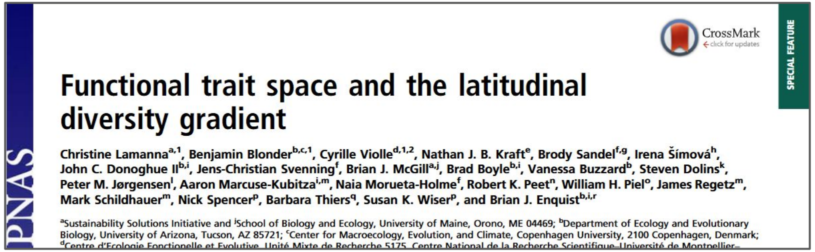

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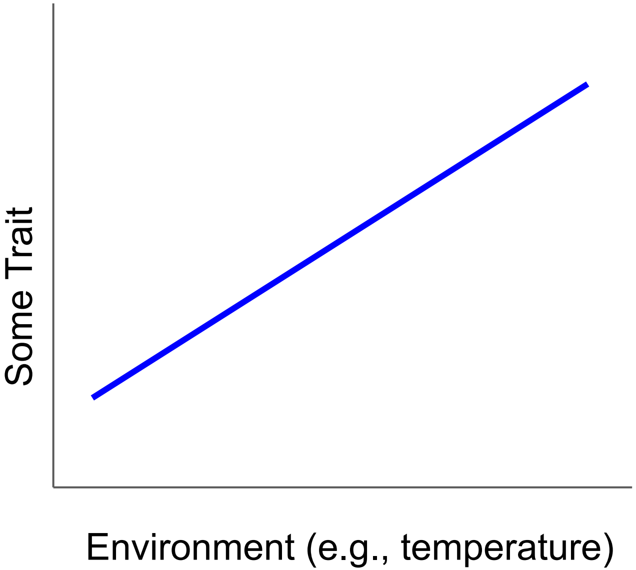
Trait-environment relationships



Trait-environment relationships

ARTICLES
https://doi.org/10.1038/s41559-021-01396-1
nature ecology & evolution
Check for updates

Consistent trait–environment relationships within and across tundra plant communities

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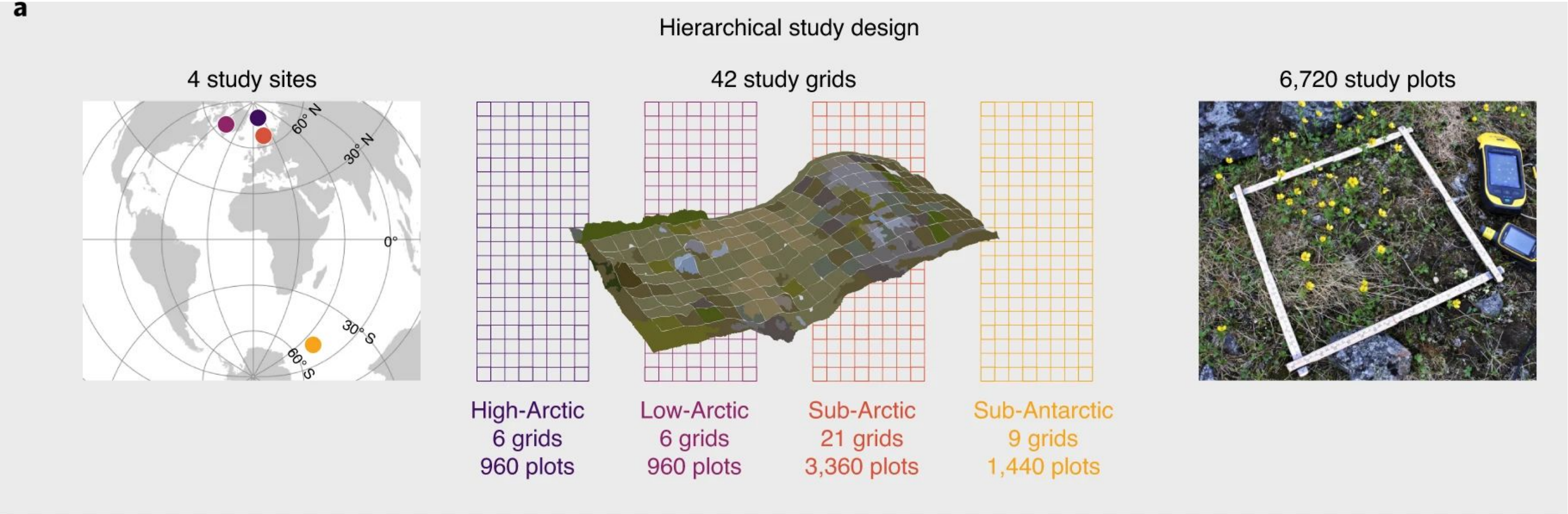

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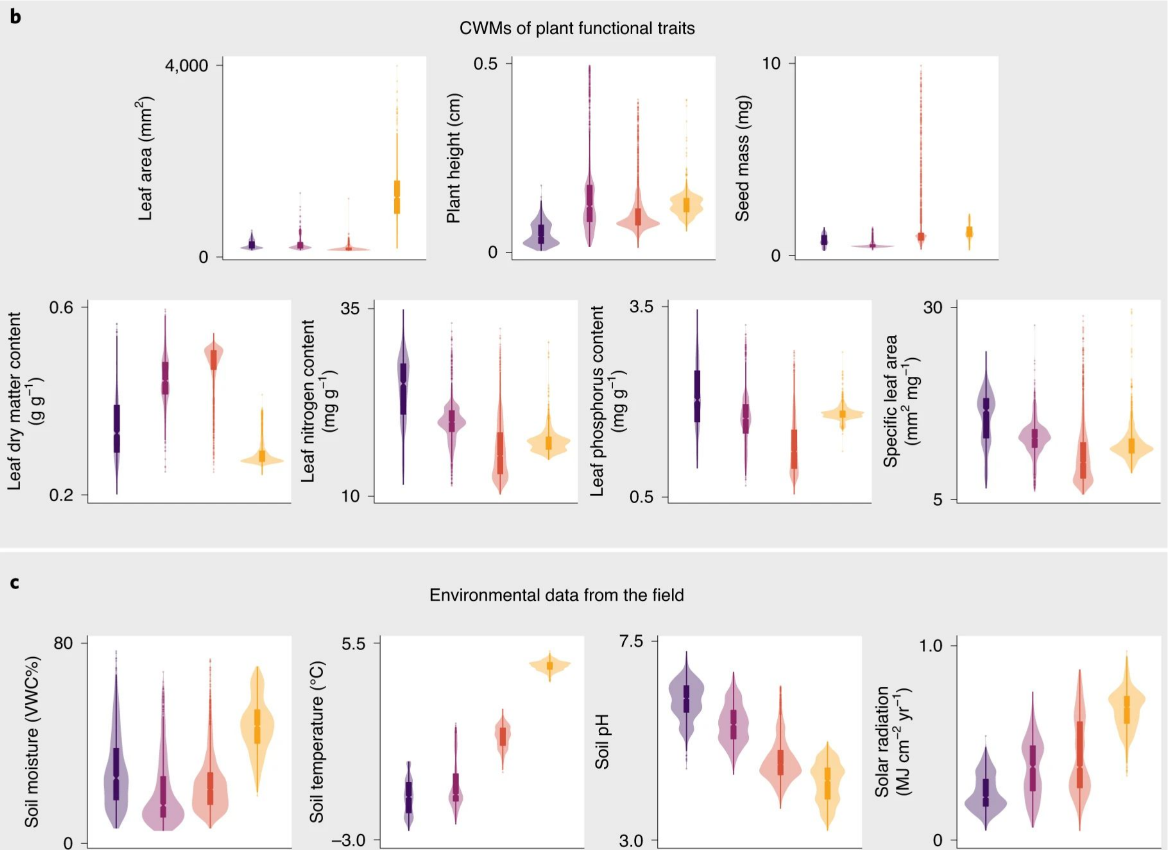

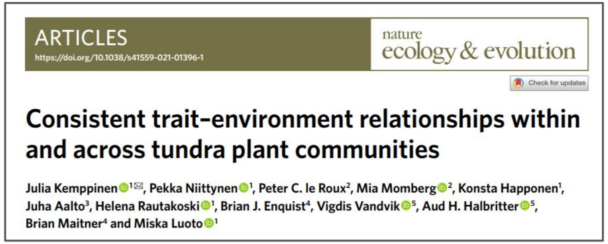
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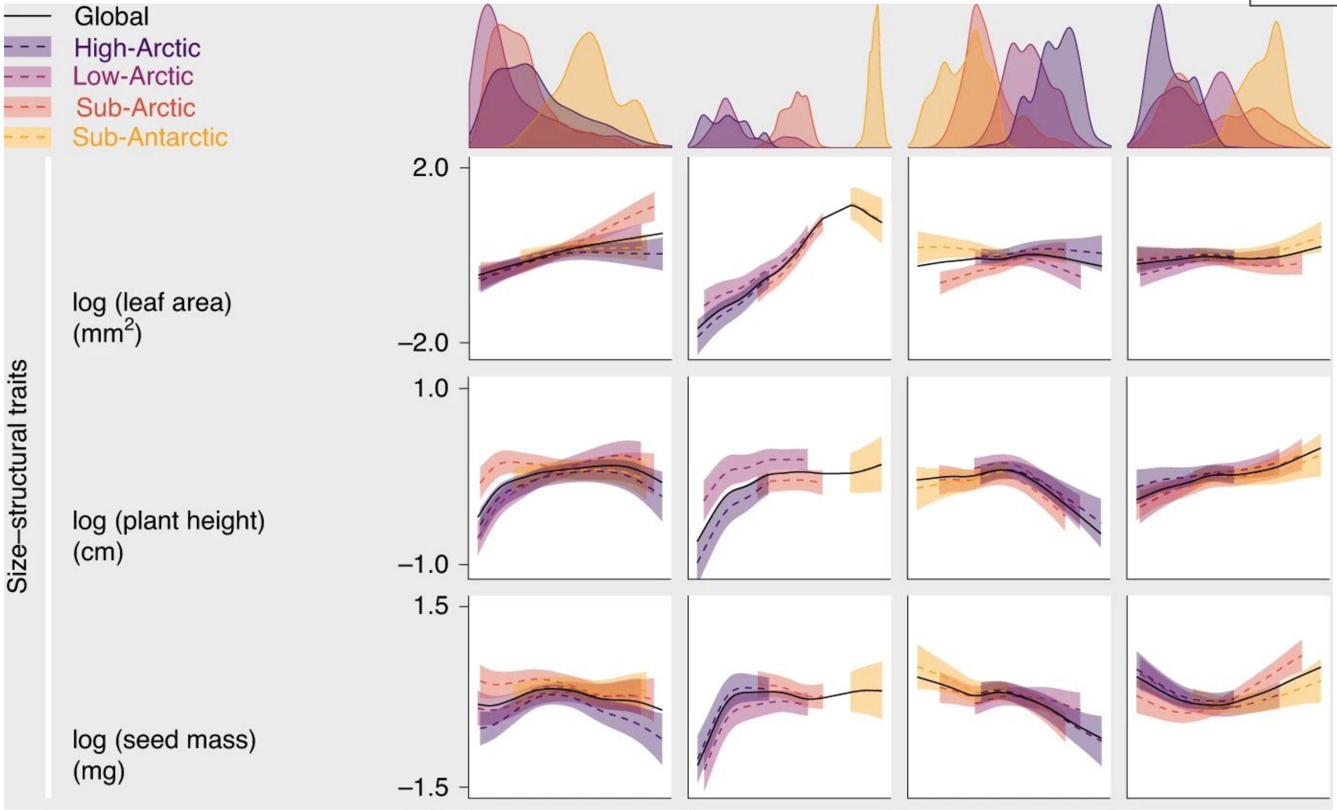

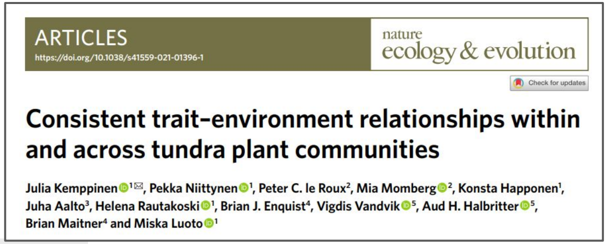

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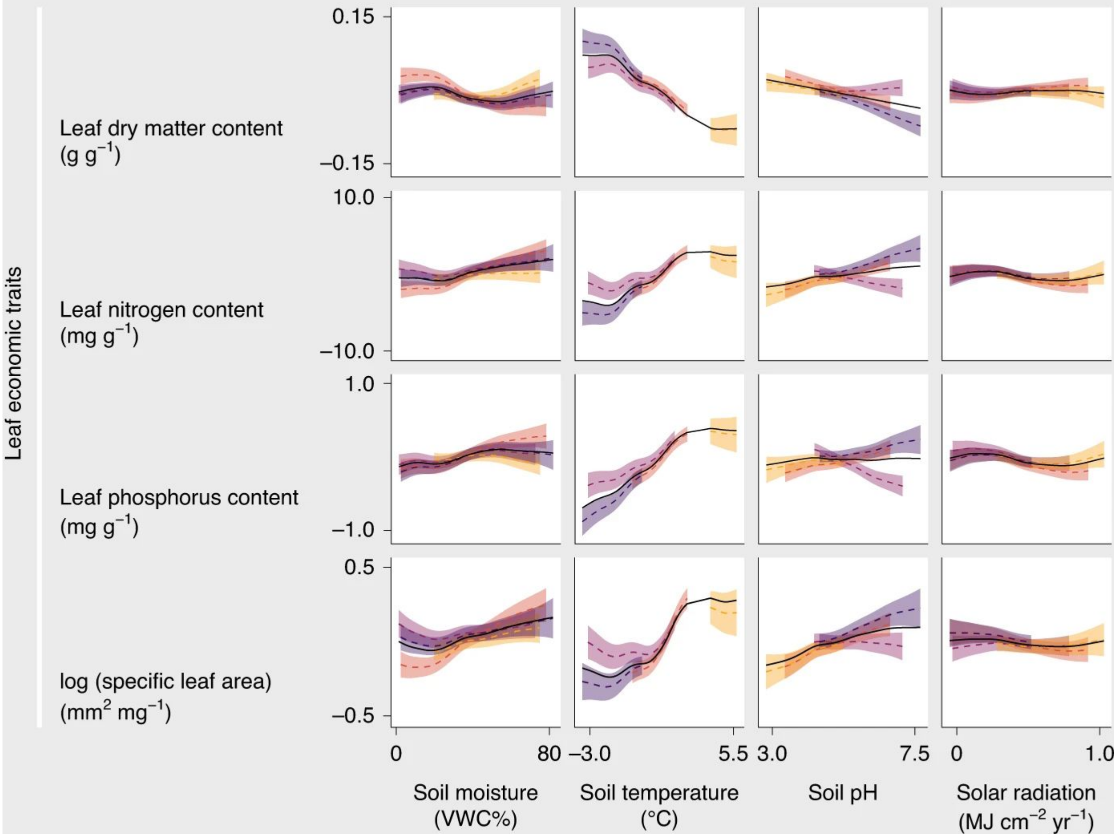

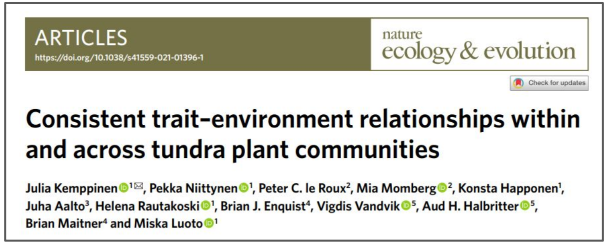

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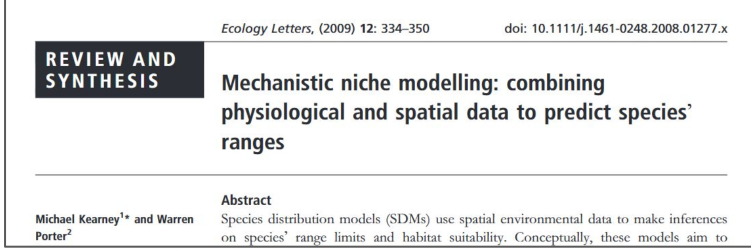
Physiological models

We can use theory to develop models that explicitly link traits to performance

Challenging and data intense, but more mechanistic

Physiological models



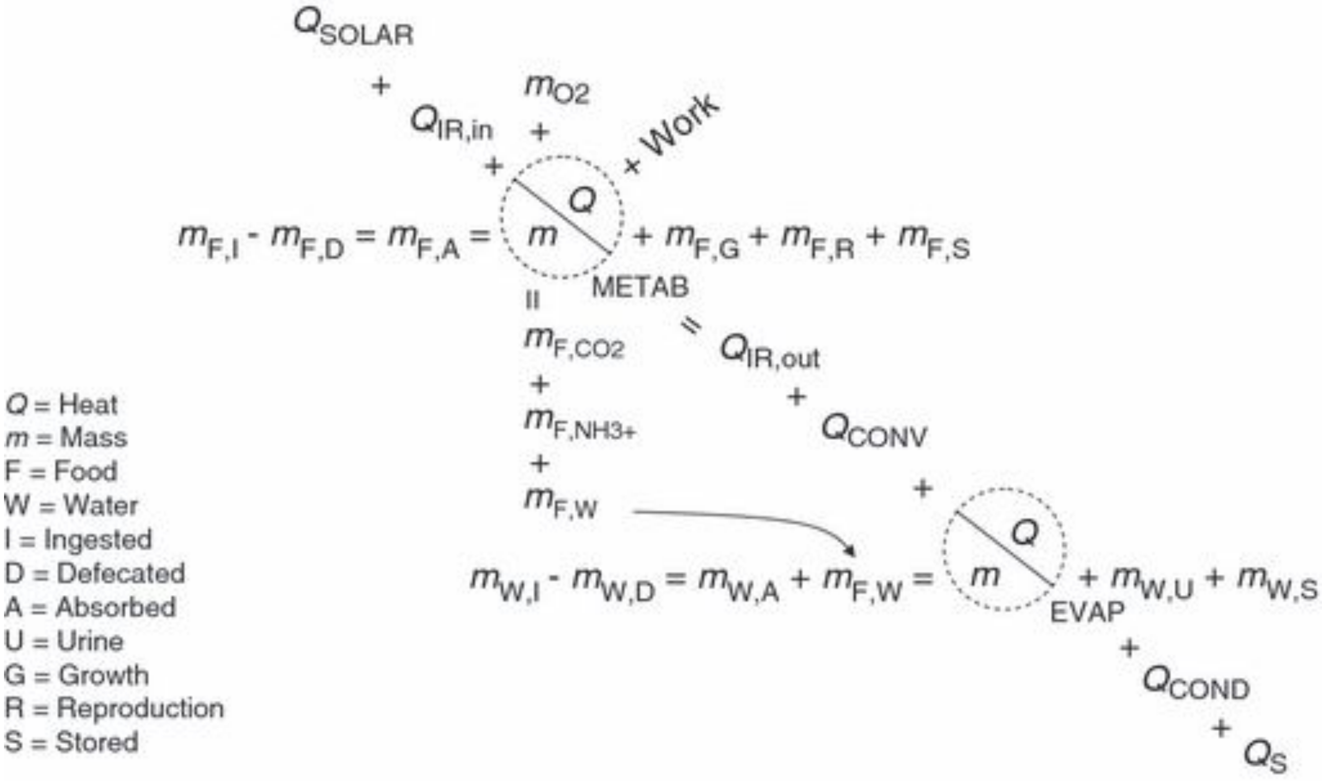
Ecology Letters, (2009) **12**: 334–350 doi: 10.1111/j.1461-0248.2008.01277.x

REVIEW AND SYNTHESIS

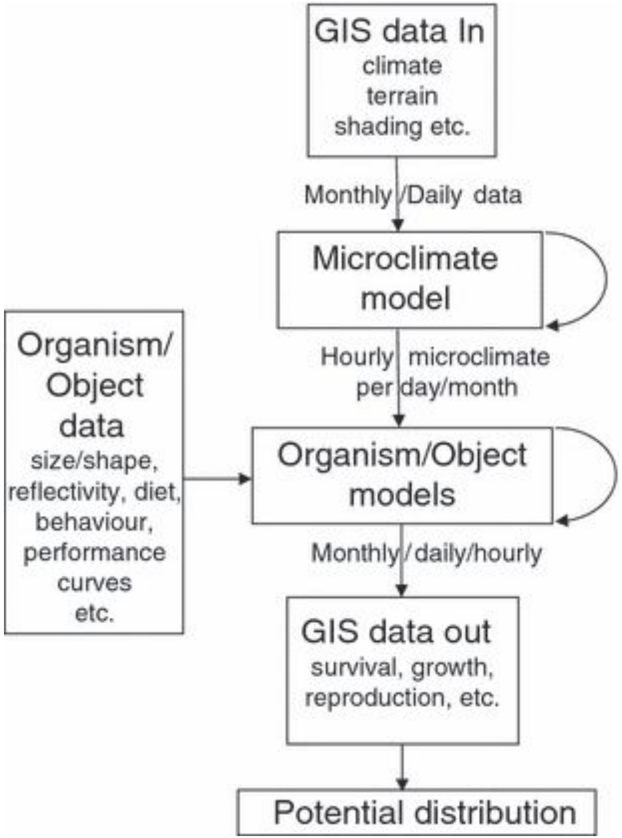
Mechanistic niche modelling: combining physiological and spatial data to predict species' ranges

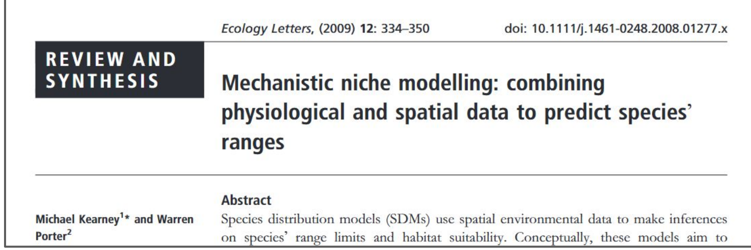
Michael Kearney[1]* and Warren Porter[2]

Abstract
Species distribution models (SDMs) use spatial environmental data to make inferences on species' range limits and habitat suitability. Conceptually, these models aim to



Physiological models





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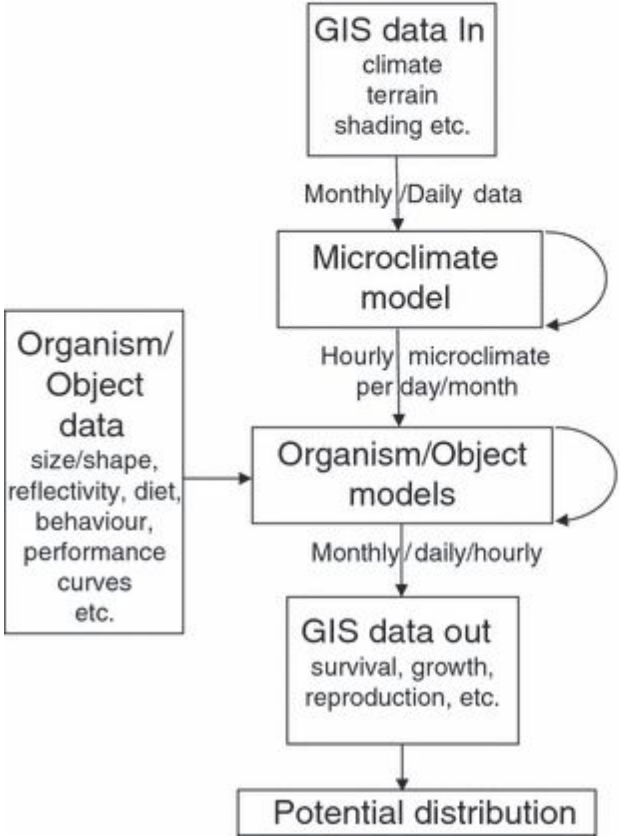

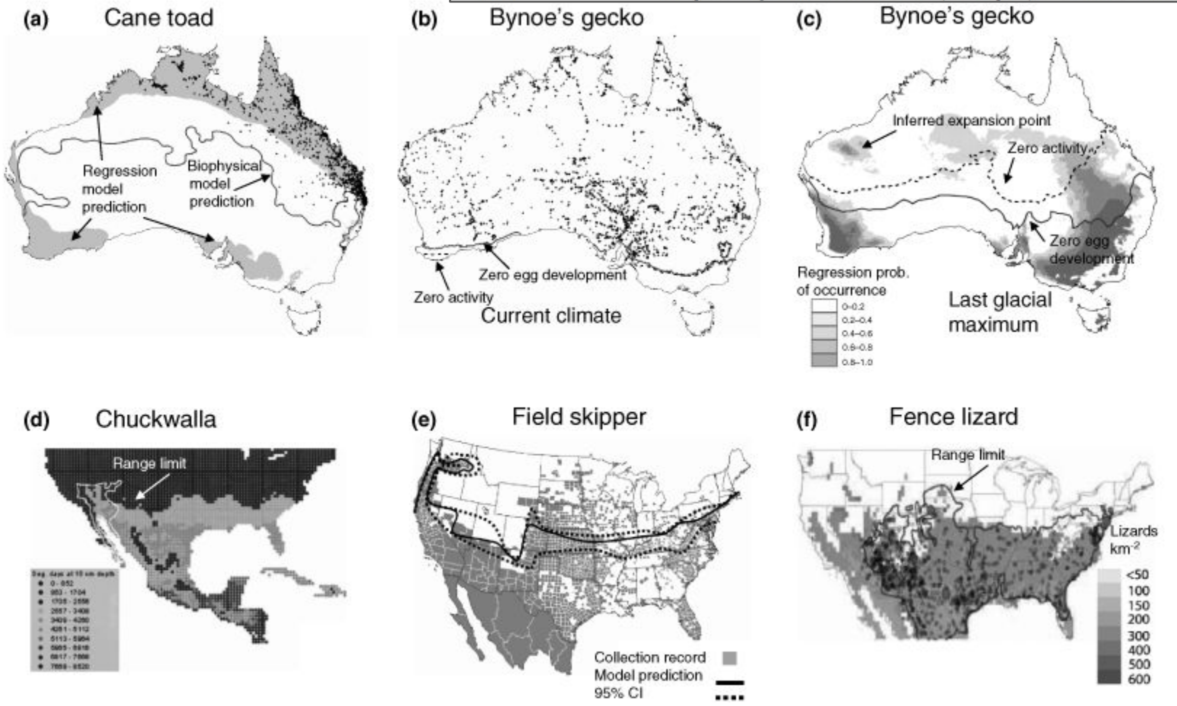

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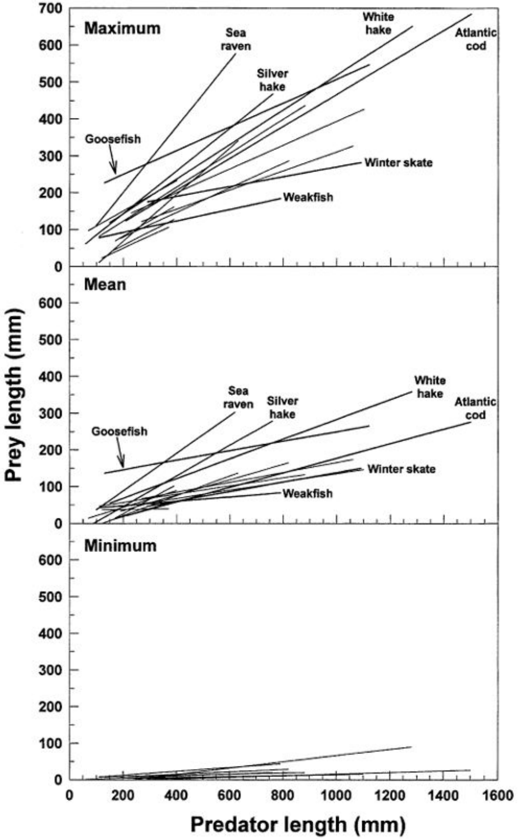
Traits and biotic relationships

- Competition
- Predation
- Mutualisms

Traits and Predation



Vol. 208: 229–248, 2000 | MARINE ECOLOGY PROGRESS SERIES Mar Ecol Prog Ser | Published December 8

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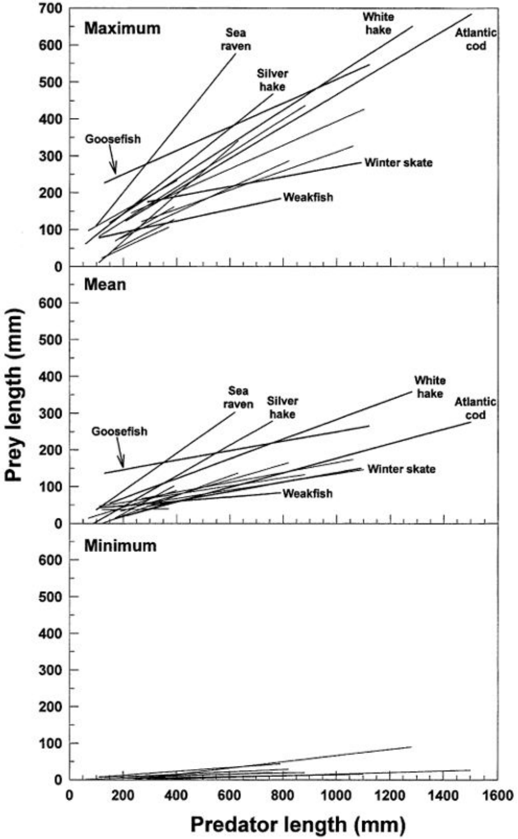

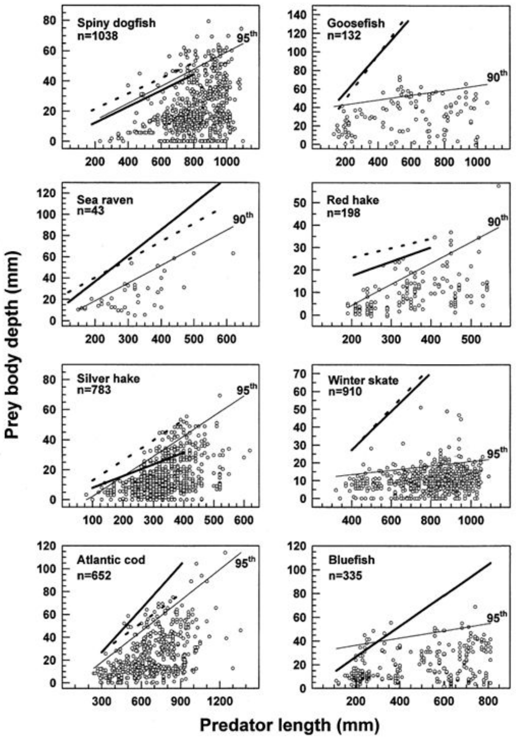

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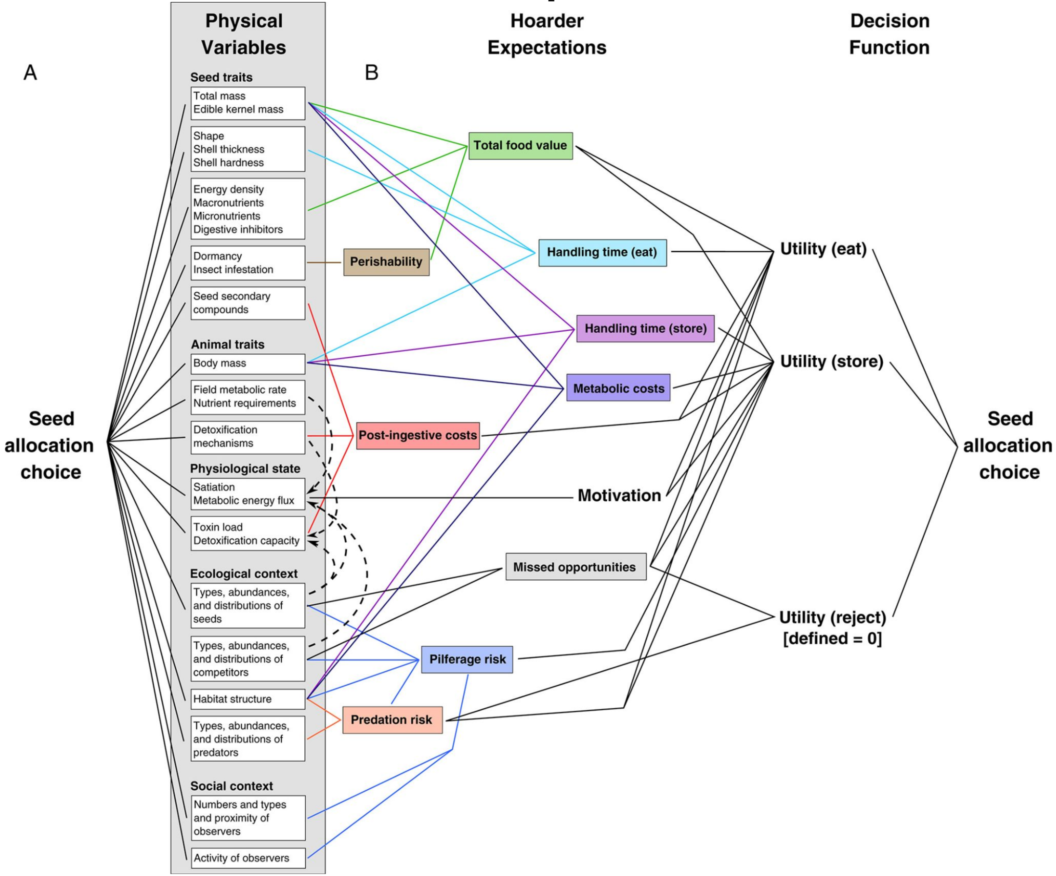
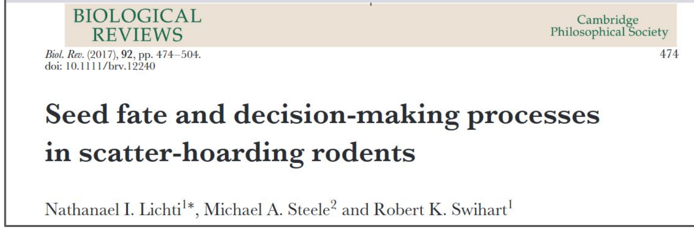
Solid = Pred gape width

Dashed = Pred gape height

Solid = Max prey body depth v predator body size

Traits and seed predation





BIOLOGICAL REVIEWS
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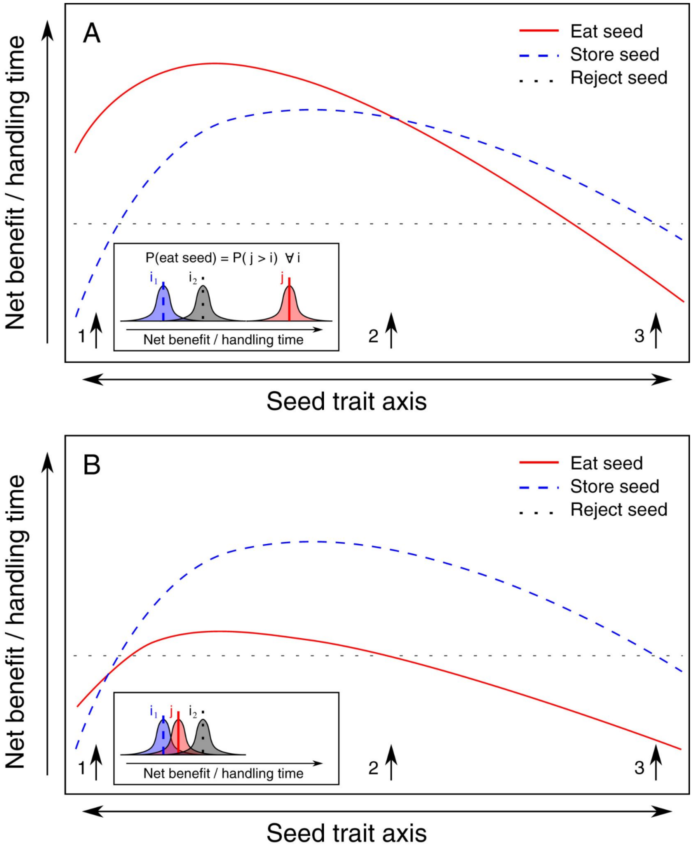
Seed fate and decision-making processes in scatter-hoarding rodents

Nathanael I. Lichti[1]*, Michael A. Steele[2] and Robert K. Swihart[1]

Traits and seed predation



Hungry

Full

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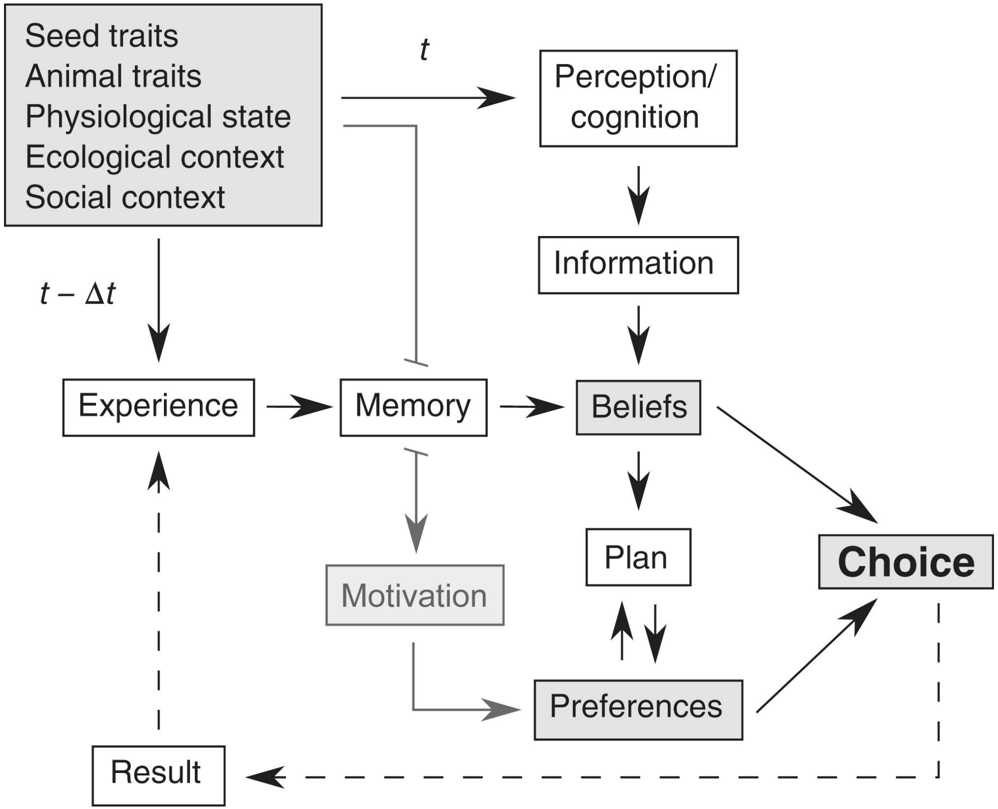

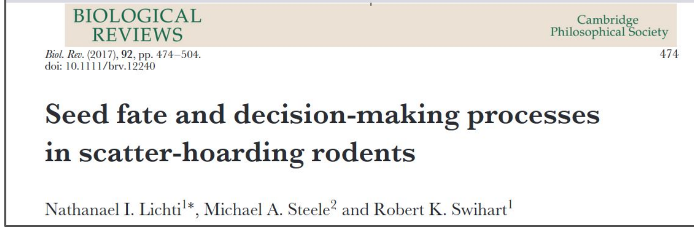
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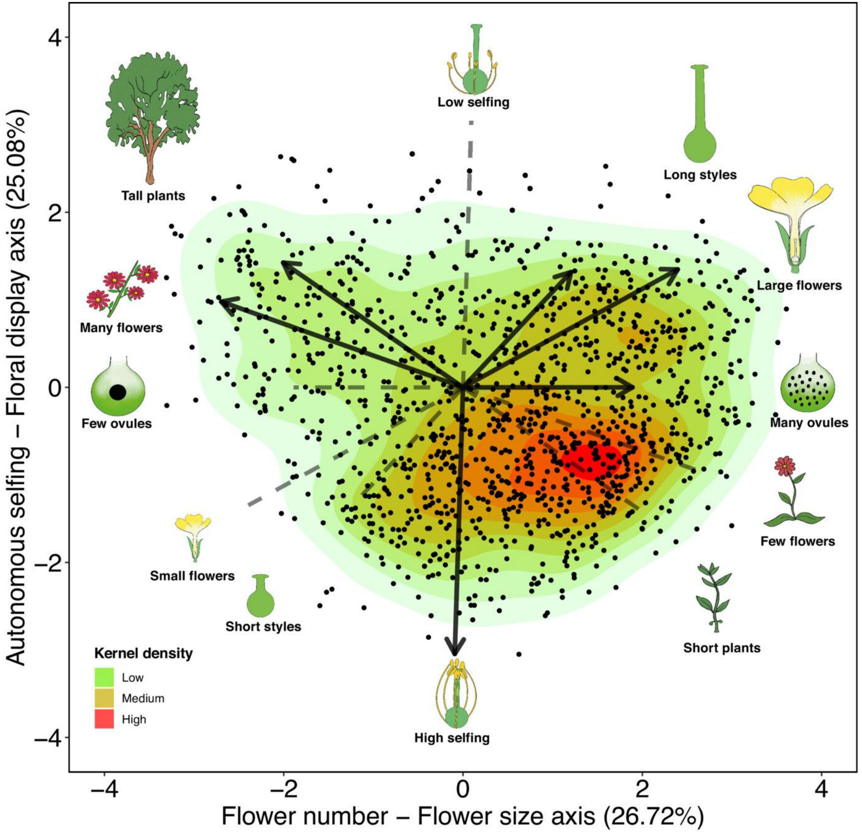

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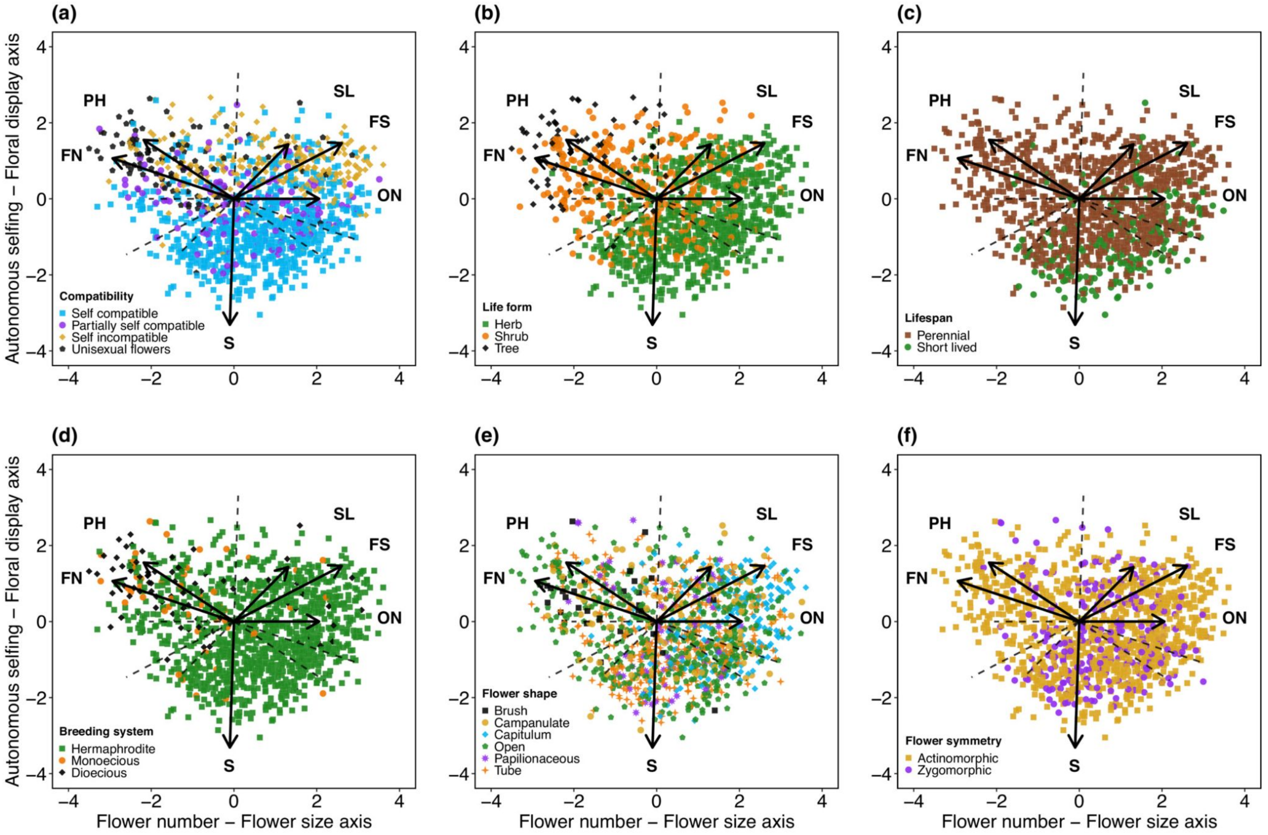
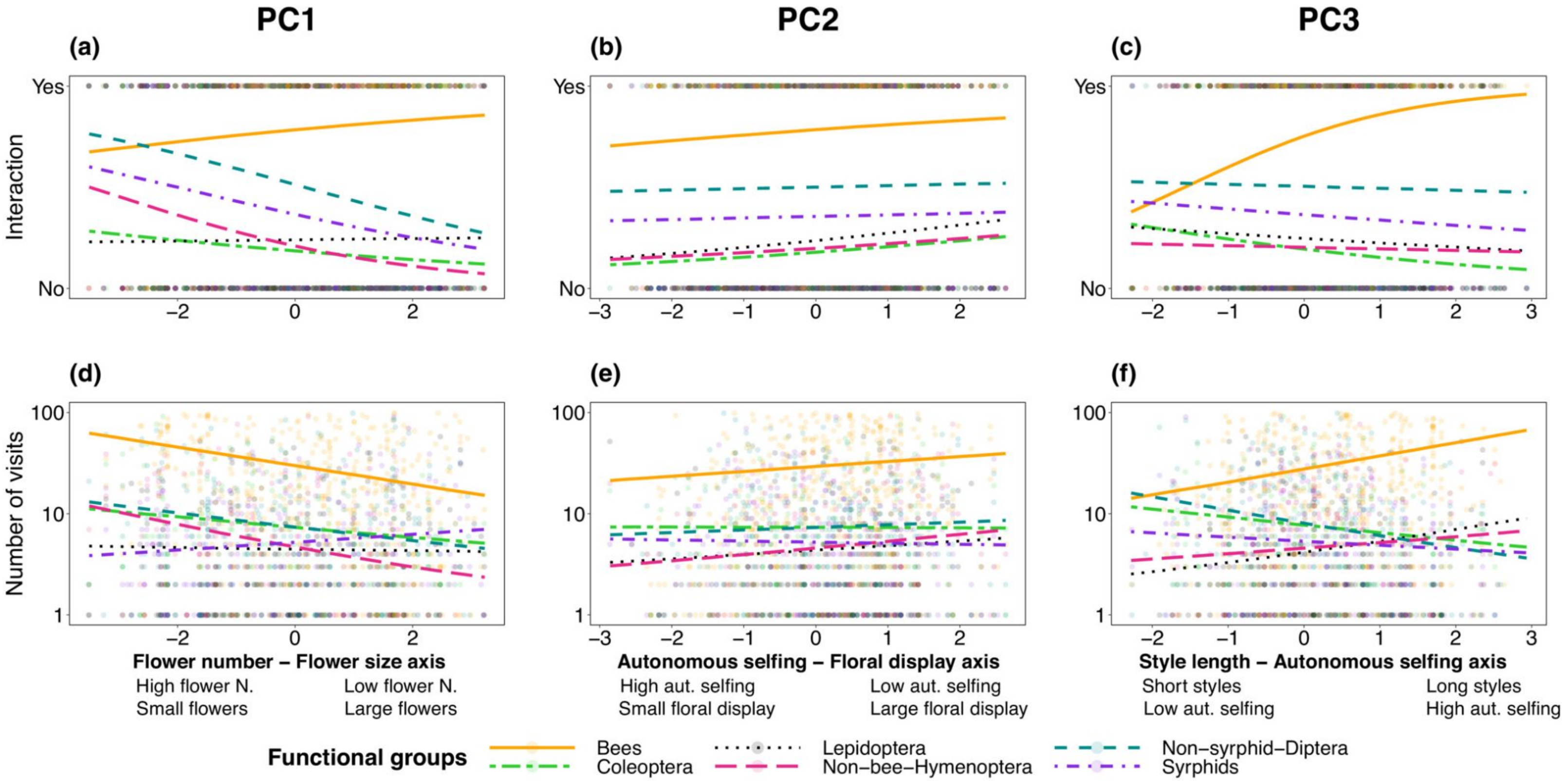
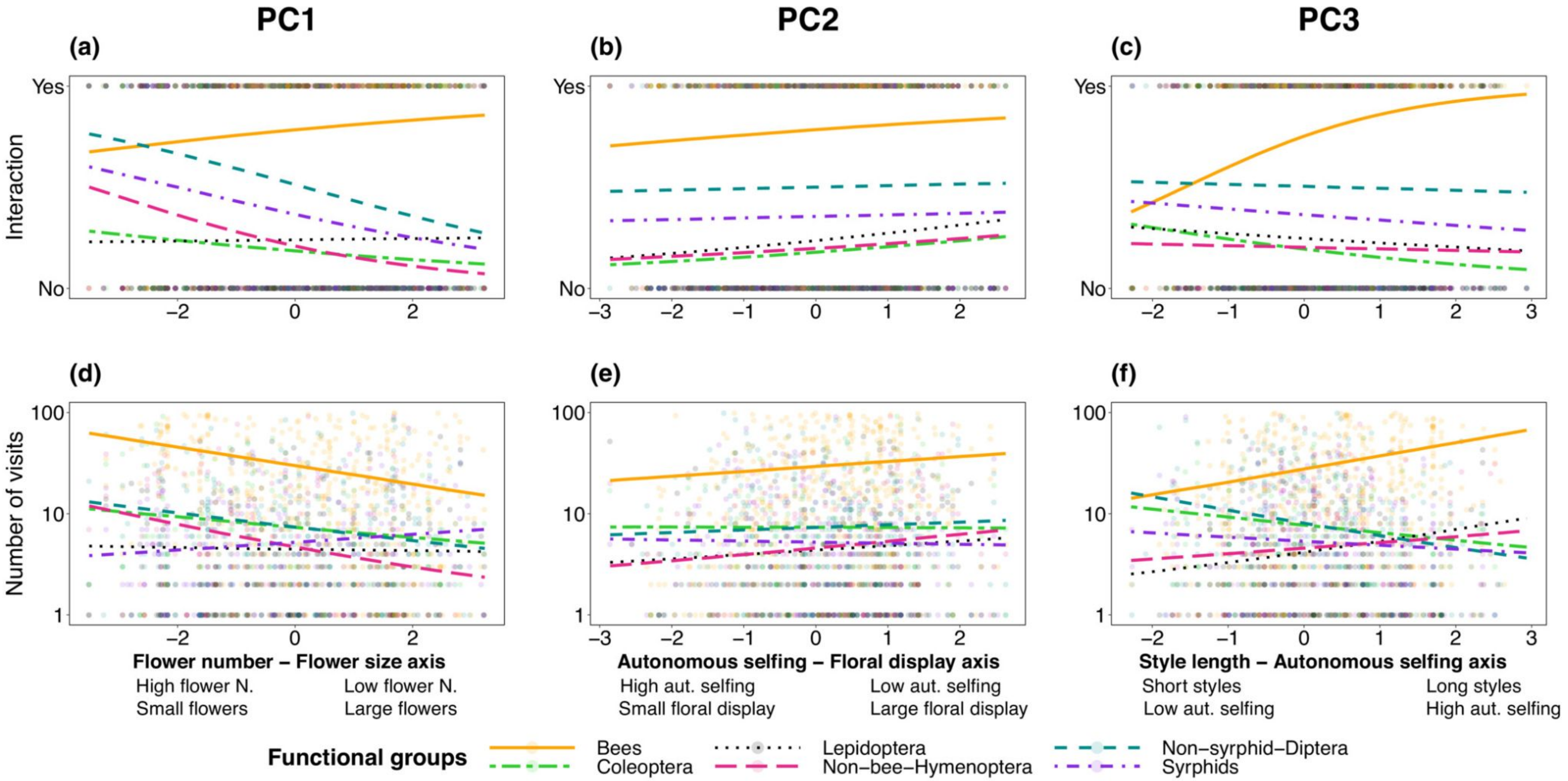


Traits and mutualism



Traits and mutualism

How might we improve our ability to predict these which pollinators interact with which plant?

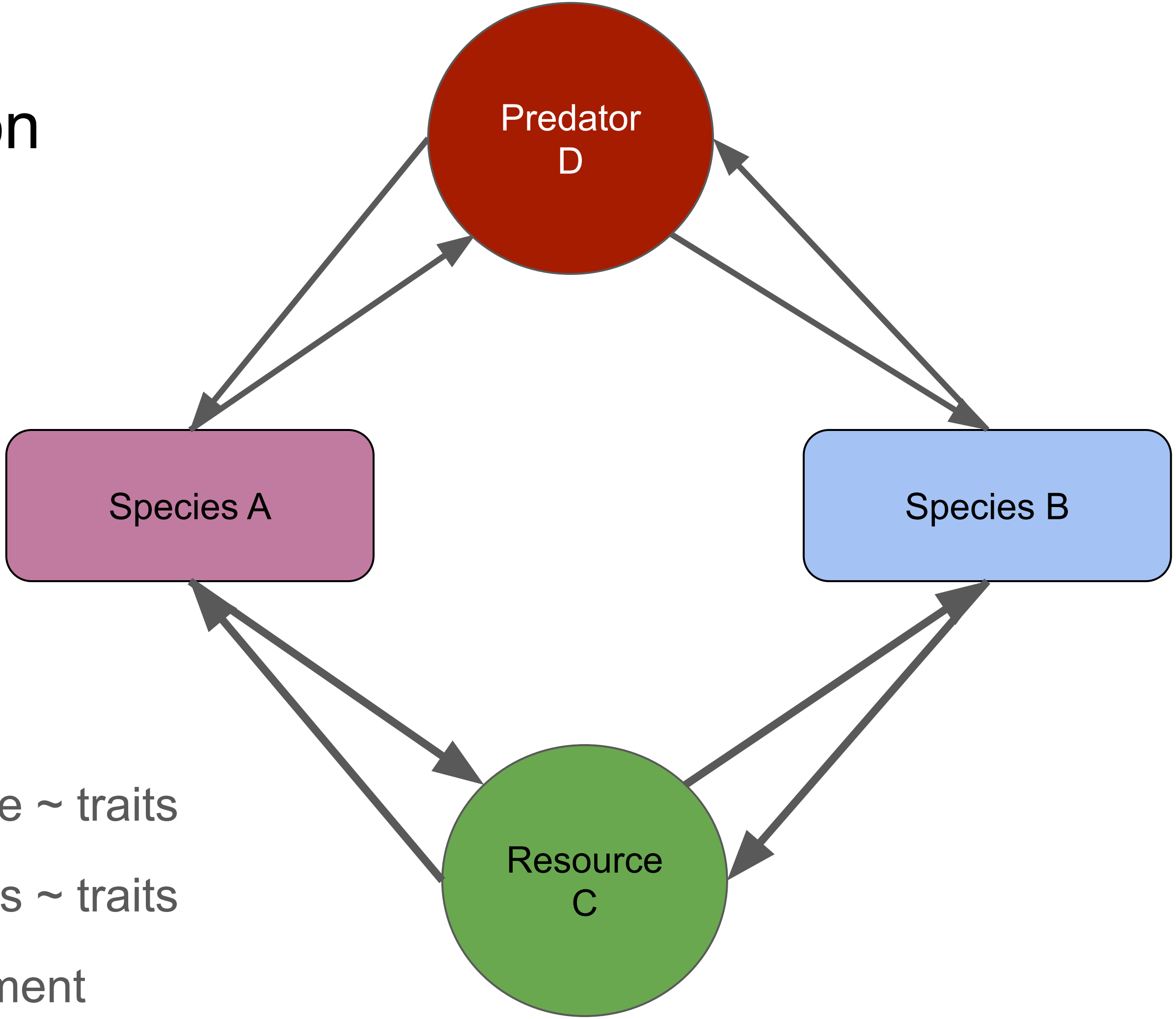


Traits and competition

- Notoriously difficult

Traits and competition

- Notoriously difficult



Remember, you need:

Impact of species on resource ~ traits

Impact of resource on species ~ traits

These can vary with environment

Traits and competition

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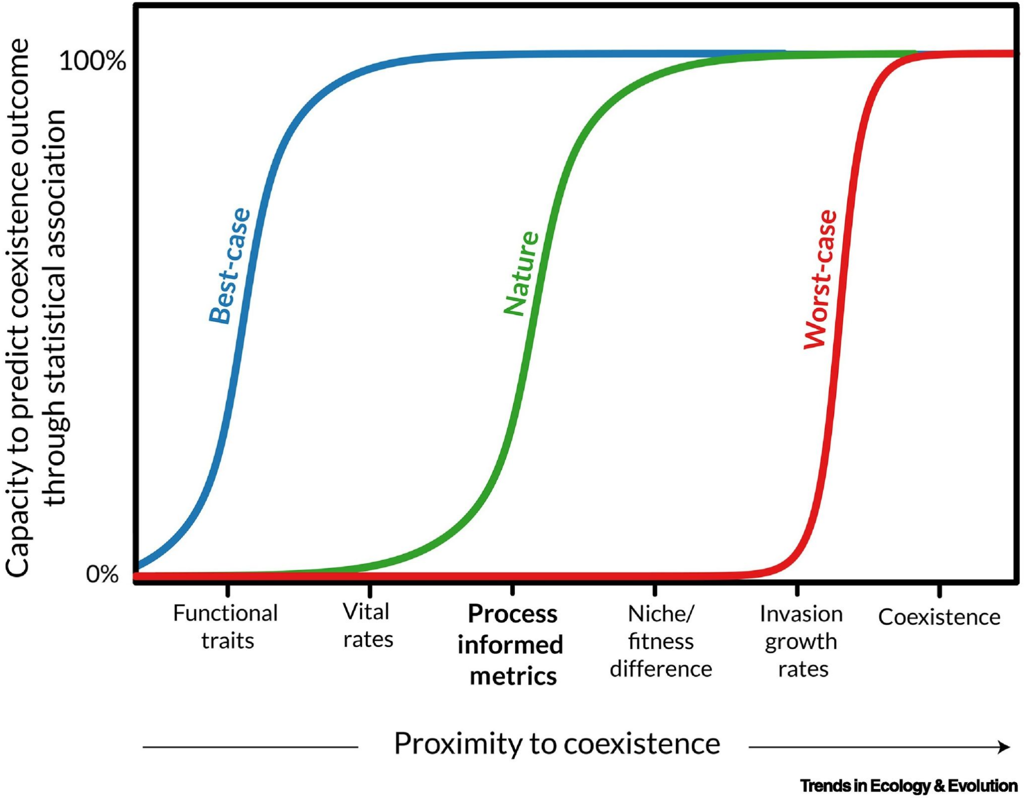
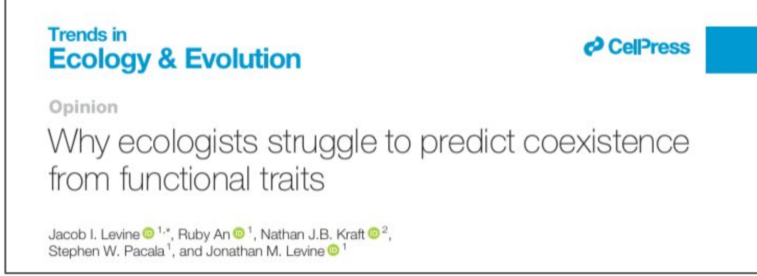
Opinion

Why ecologists struggle to predict coexistence from functional traits

Jacob I. Levine[1,*], Ruby An[1], Nathan J.B. Kraft[2], Stephen W. Pacala[1], and Jonathan M. Levine[1]

Traits and competition

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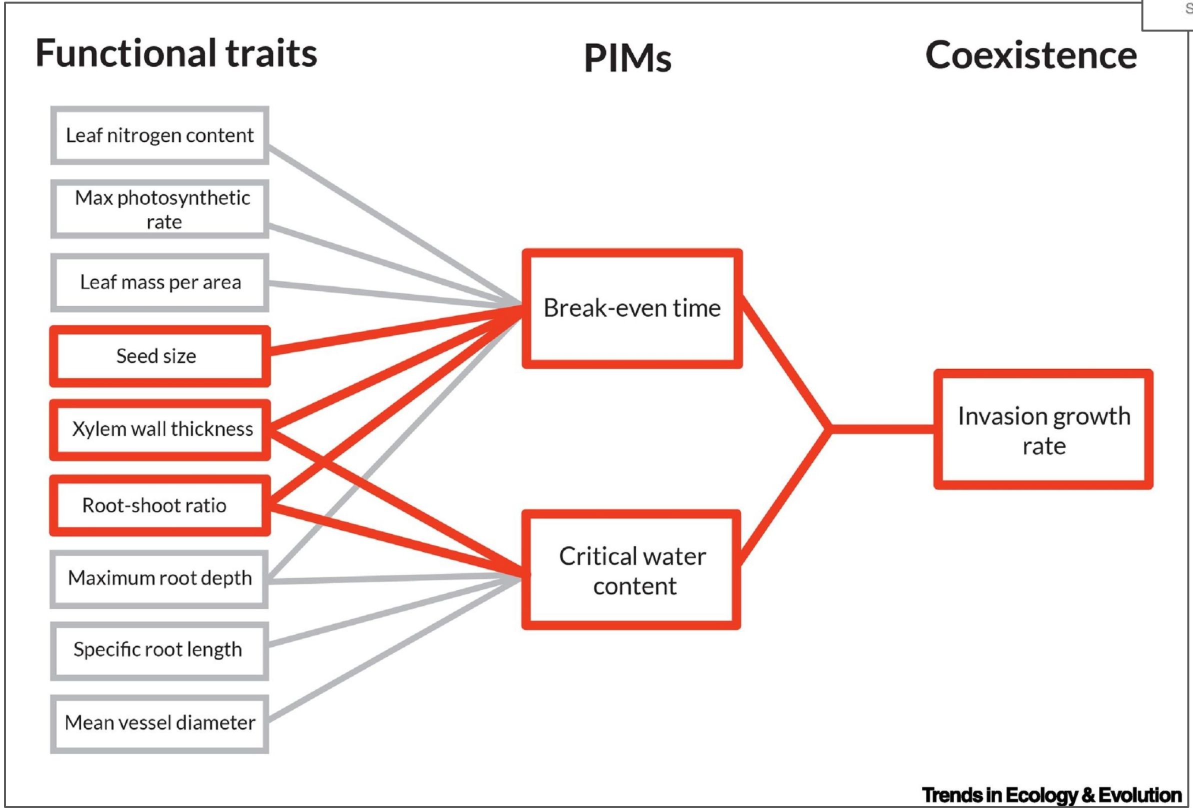

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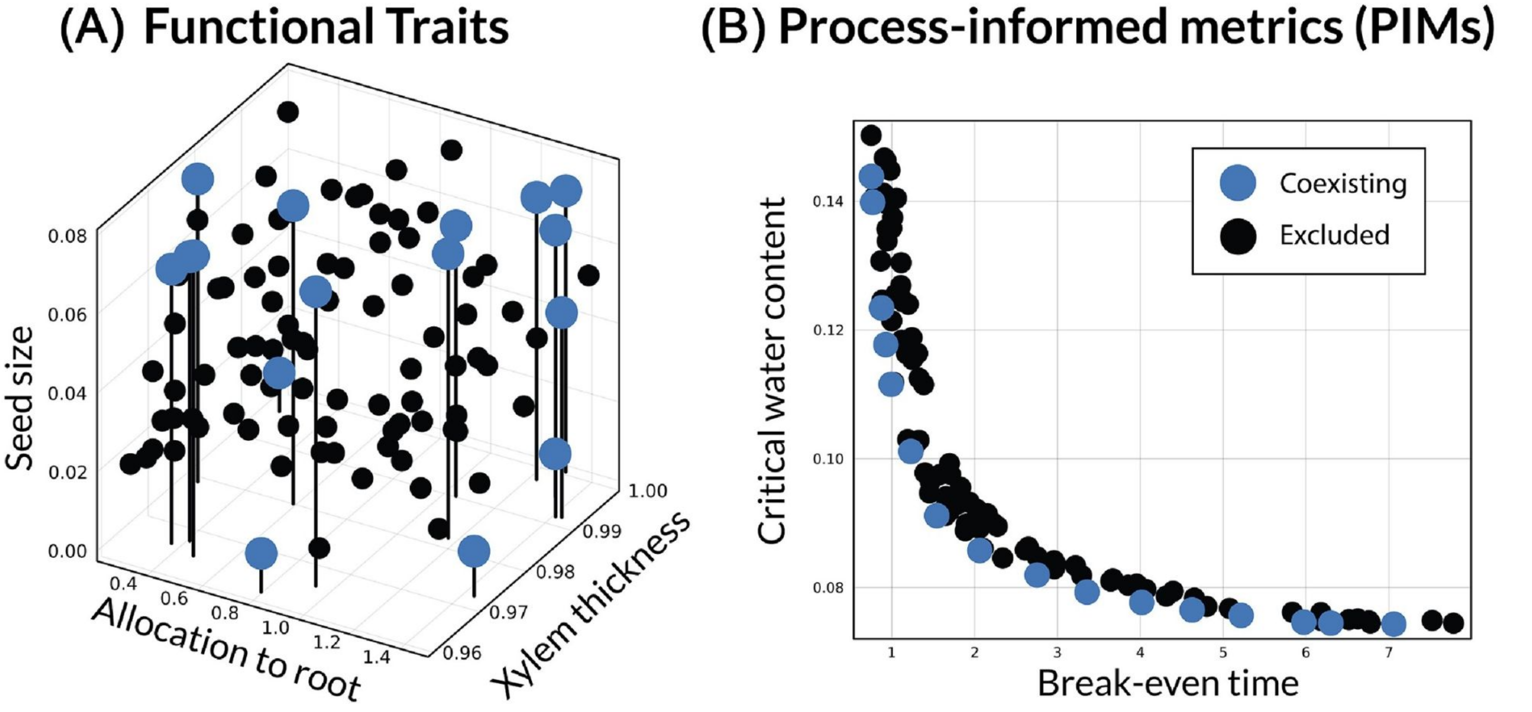

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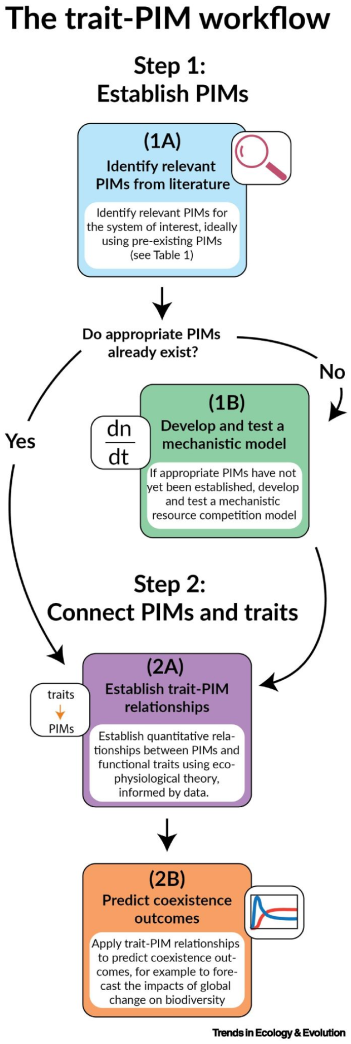
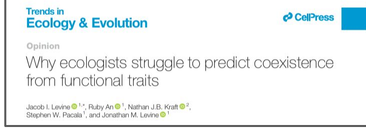
Traits and competition





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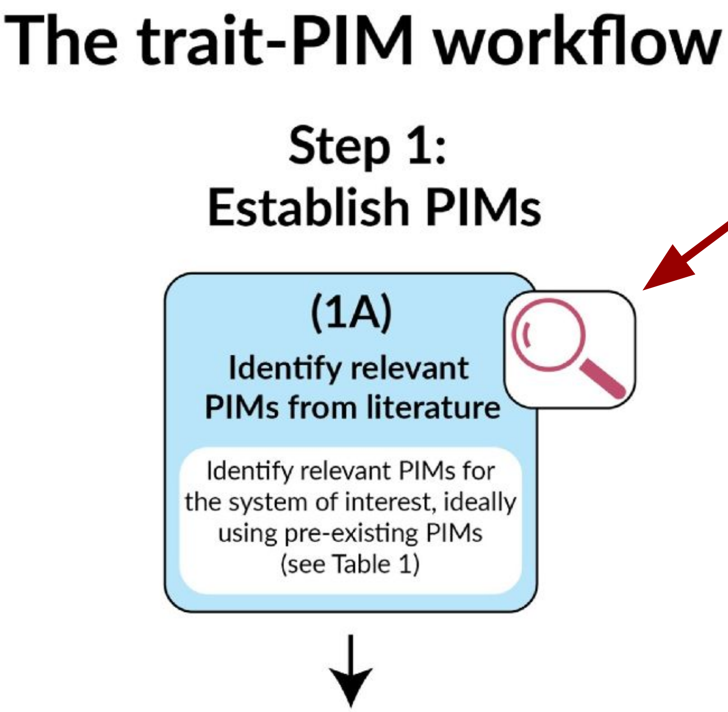

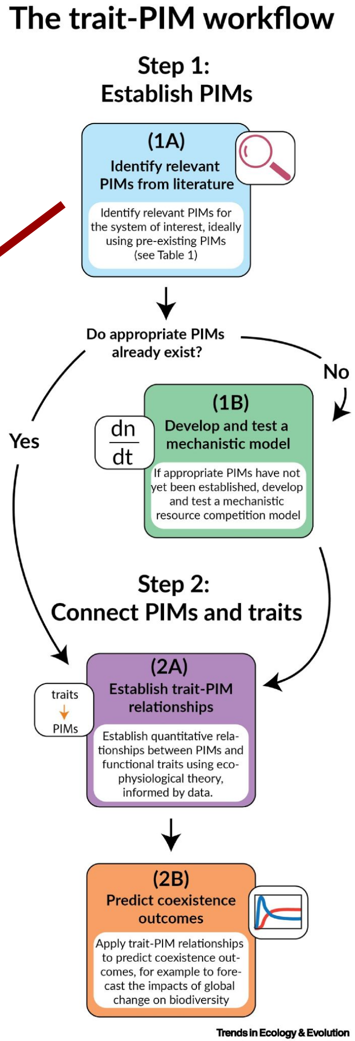

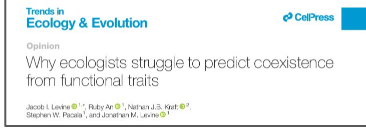
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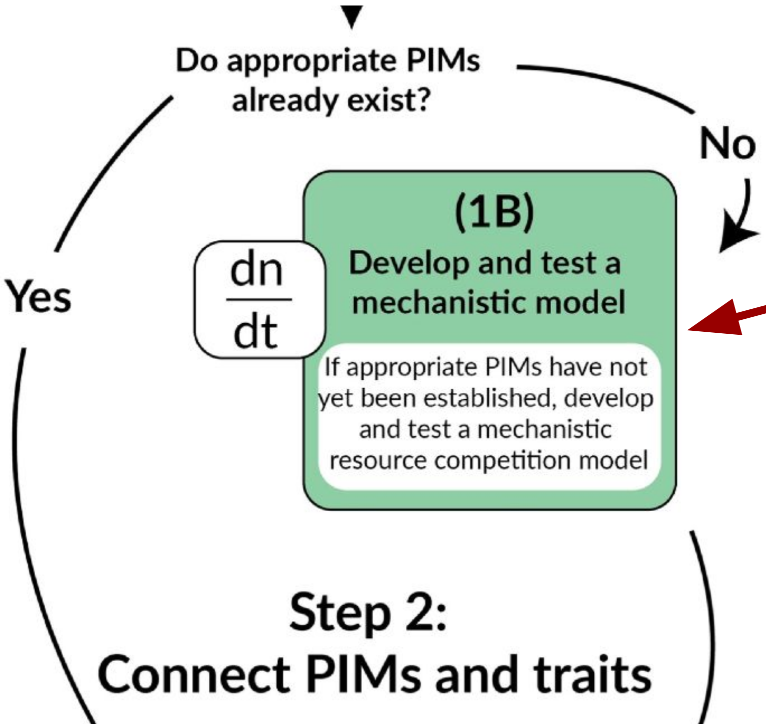

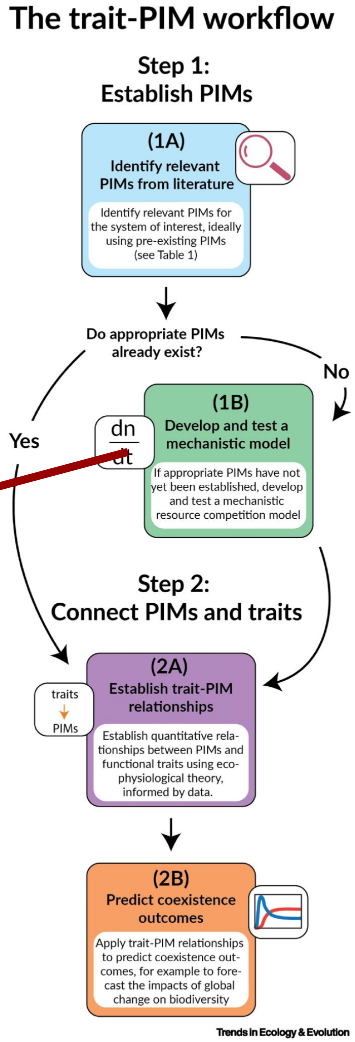


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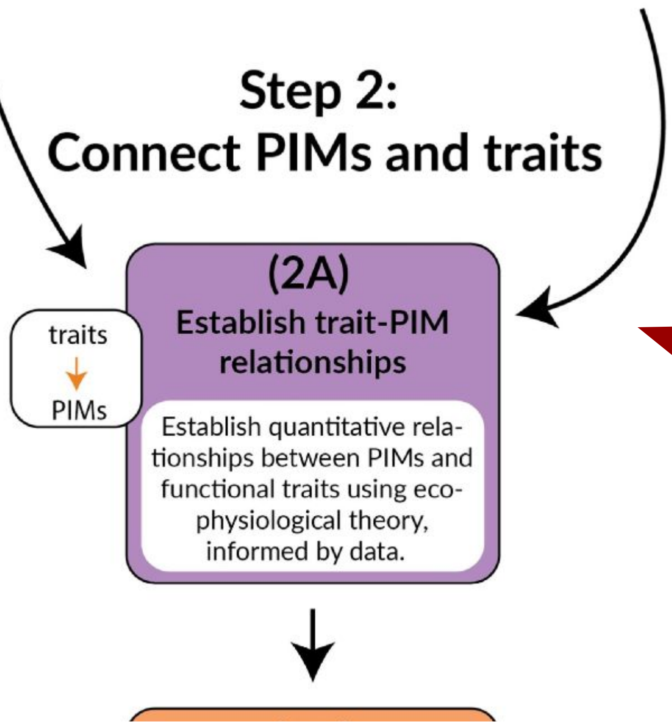

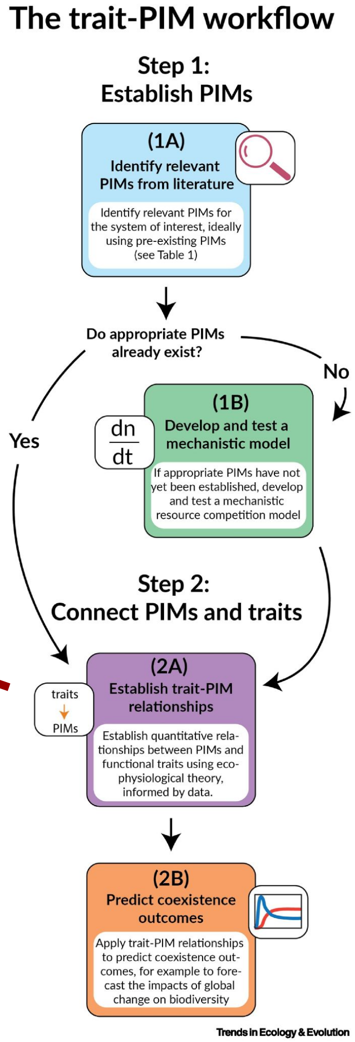


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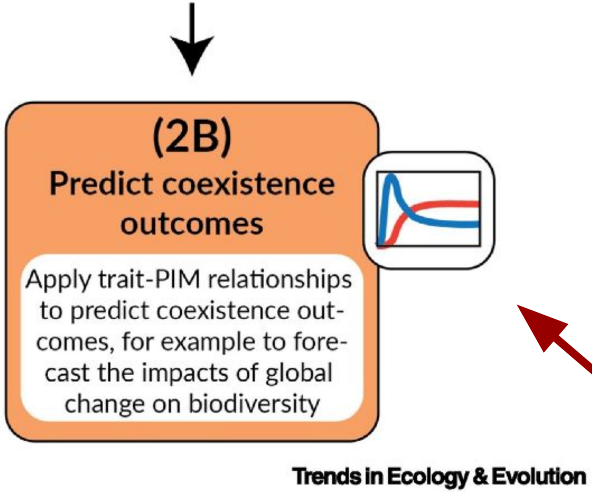
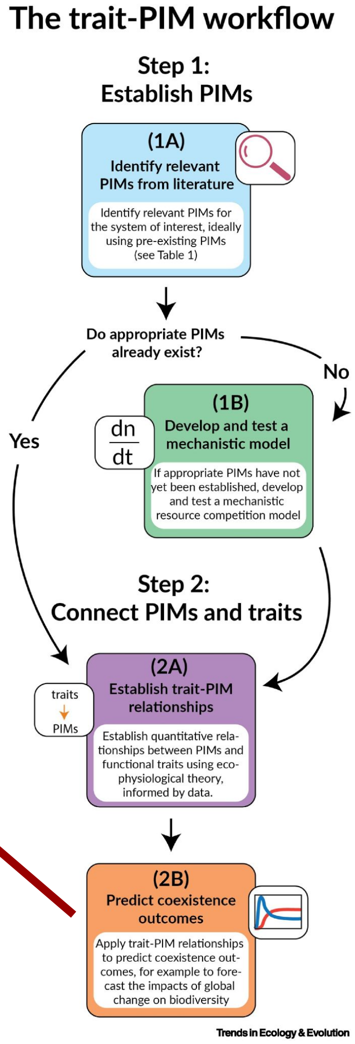




Traits and community assembly

Abiotic filtering

Biotic filtering

Dispersal

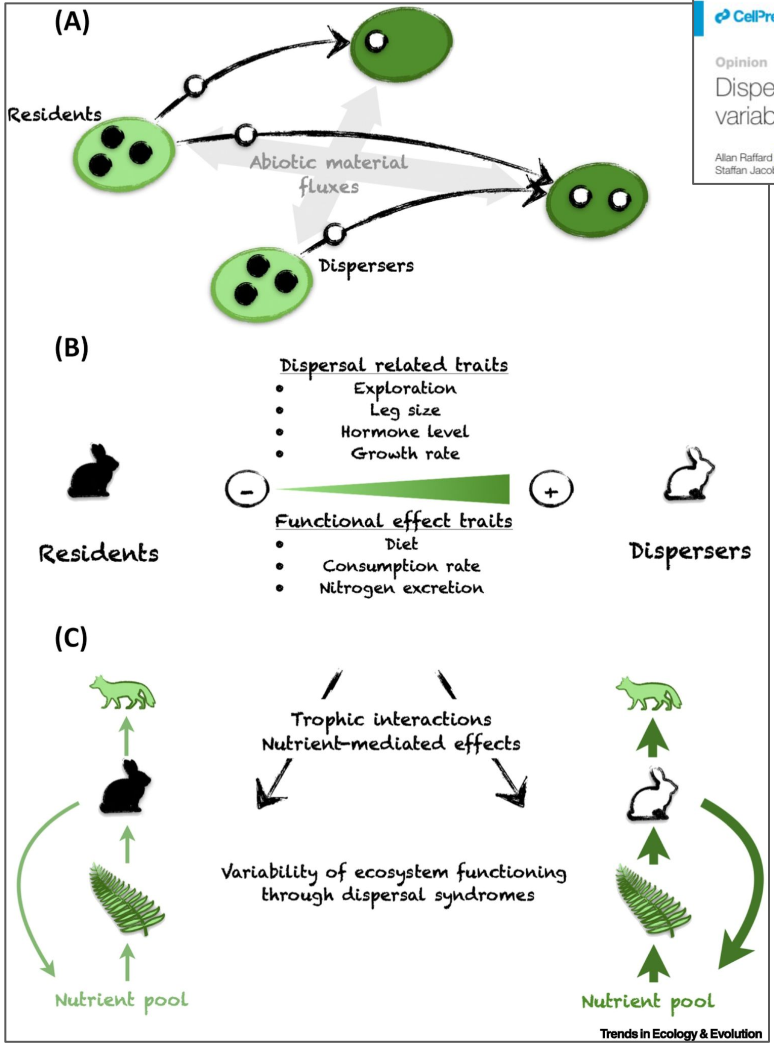
Traits and community assembly

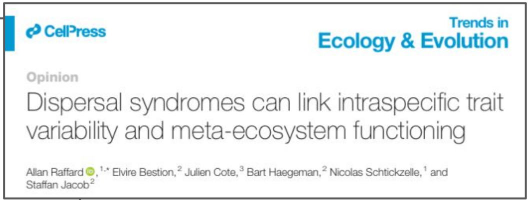
Abiotic filtering

Biotic filtering

$\boxed{\text{Dispersal}}$

Traits and dispersal

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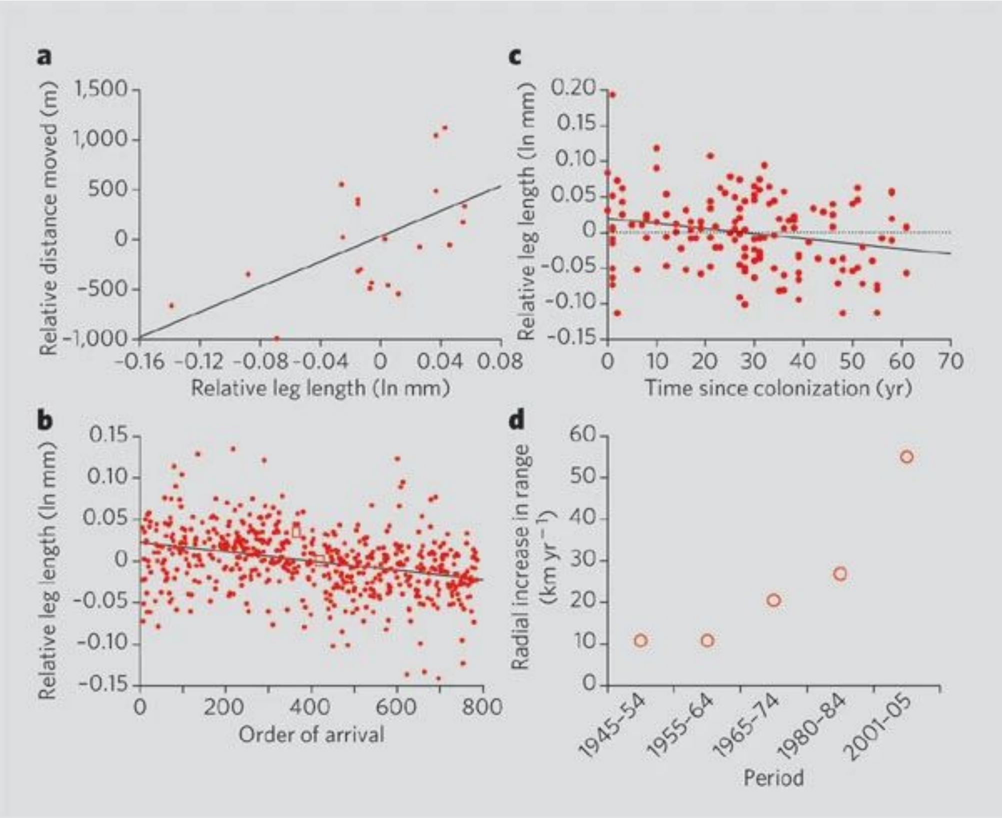
Dispersal syndromes can link intraspecific trait variability and meta-ecosystem functioning

Allan Raffard,[1,*] Elvire Bestion,[2] Julien Cote,[3] Bart Haegeman,[2] Nicolas Schtickzelle,[1] and Staffan Jacob[2]

Traits and dispersal



Vol 439|16 February 2006 nature

BRIEF COMMUNICATIONS

Invasion and the evolution of speed in toads

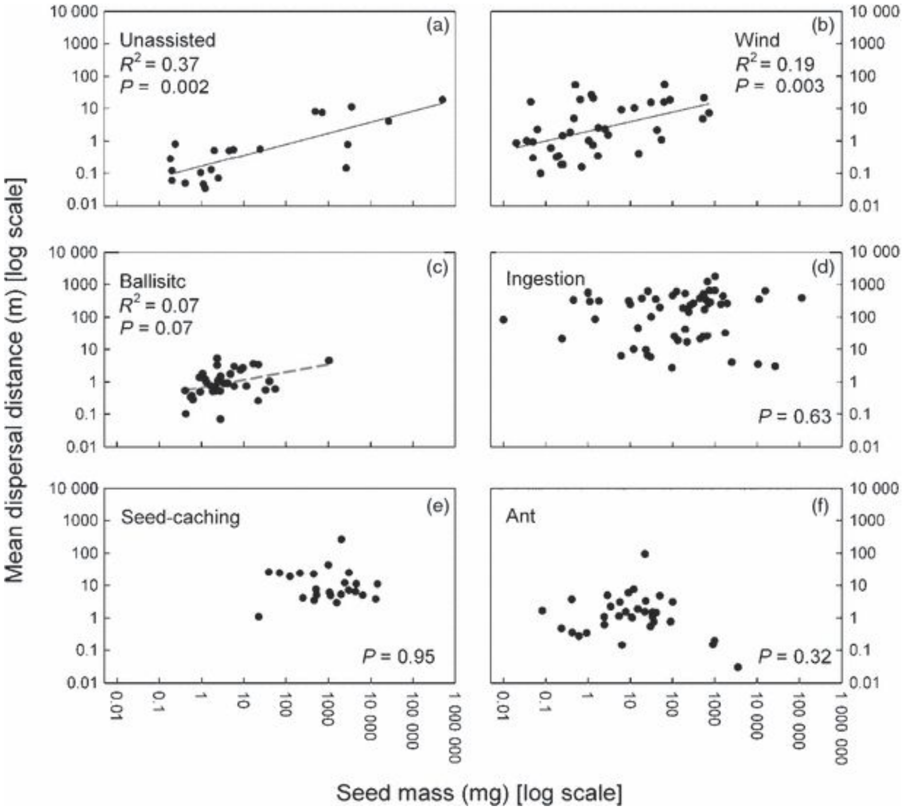
Cane toads seem to have honed their dispersal ability to devastating effect over the generations.

Traits and seed dispersal

Journal of Ecology
British Ecological Society
Journal of Ecology 2011, **99**, 1299–1307 doi: 10.1111/j.1365-2745.2011.01867.x

Seed dispersal distance is more strongly correlated with plant height than with seed mass

Fiona J. Thomson[1,2]*, Angela T. Moles[2], Tony D. Auld[3] and Richard T. Kingsford[1]



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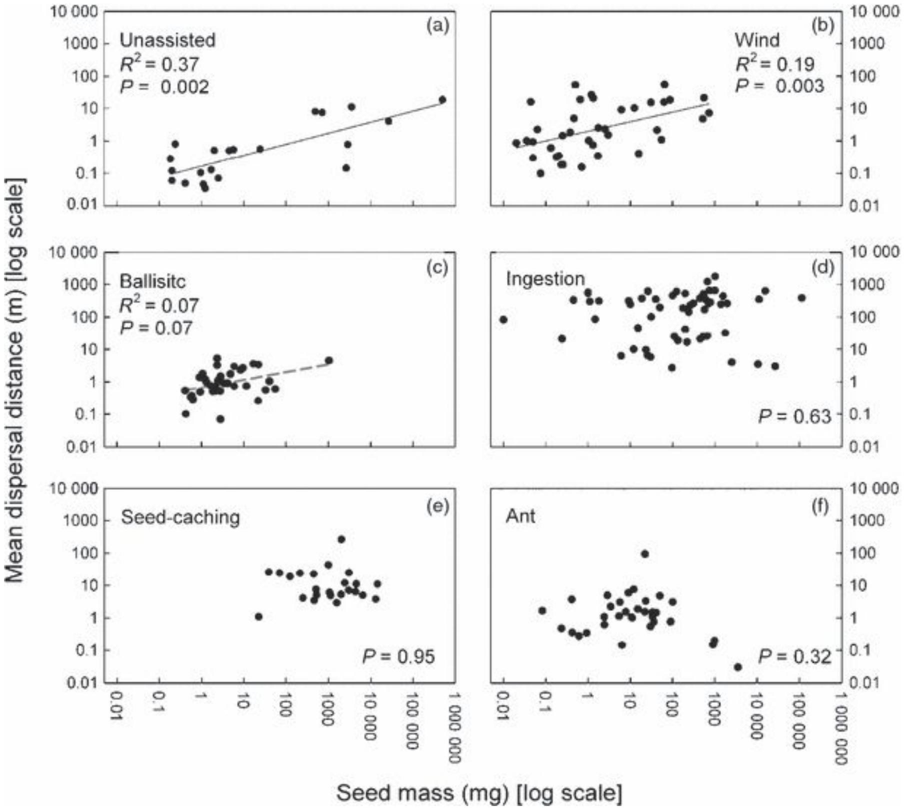

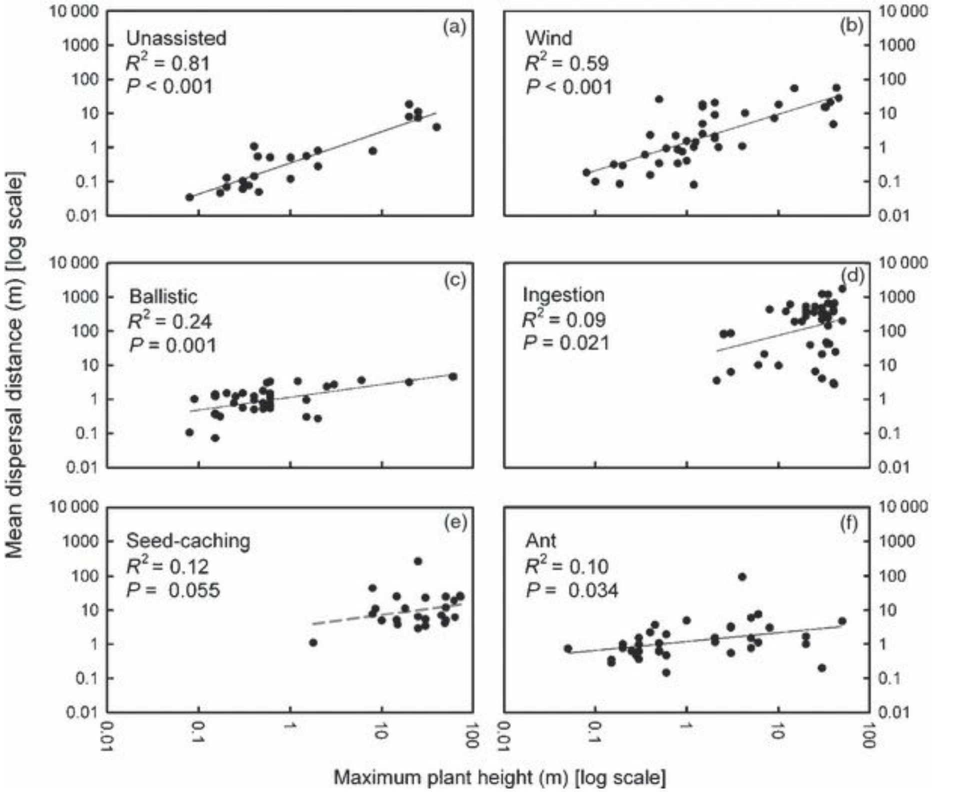

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Why might plant height predict dispersal distance?

Traits and seed dispersal

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Why might plant height predict dispersal distance?

- Small species rely on seedbank, dispersing through time vs space
- Big species, more natural enemies, more need to get away
- Falling seeds drift further (further to fall)
- Big plants, big fruit, bigger dispersers which travel further?

Integrating processes

- Remember, these processes don’t actually happen in isolation
- E.g., being poorly suited to your environment might limit dispersal or competitive ability

Next class: Traits and ecosystem functioning